

# Encoding Pen Dynamics in Handwriting Images: An Incremental Study of Parkinson’s Disease Classification

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**Abstract:** We introduce a new formulation of pseudodynamic handwriting minimap signatures that maps normalized pen signals onto contact-only trajectories. Using 597 PaHaW recordings from 75 participants, we progress from static traces to isolated colour signals and RGB combinations, then examine texture descriptors, learned features and task fusion. An initial eight-task comparison identifies task-dependent gains for speed, altitude and azimuth. A subsequent evaluation selects encodings and classifier parameters entirely within training, while spatial controls test the association between signals and trace positions. Neither analysis establishes an overall dynamic advantage after multiplicity correction. On three tasks fixed from prior work, training-selected encoding with Local Phase Quantization and SVM reaches participant-level macro F1 of 0.6449, compared with 0.6143 for static images, 0.6279 for kinematic/pressure features and 0.6021 for frozen ResNet18 transfer features. The numerical advantage remains uncertain. Reducing eight tasks to three helps some pipelines but not others. Grad-CAM analysis of the earlier scratch-trained CNN concentrates on writing but agrees poorly with occlusion. The study contributes an explicit encoding technique and a controlled account of how its performance depends on task selection, retained signal information and the learning procedure.

## 1 INTRODUCTION

A digitizing tablet records the pen trajectory together with changes in speed, pressure and inclination. Encoding these measurements as colour adds information about the production of a trace while preserving its recognizable structure. We investigate whether these spatially encoded signals improve classification beyond the visible trajectory.

The application is the distinction between Parkinson’s disease (PD) and healthy controls (H) in PaHaW, a database of online handwriting and drawing tasks (Drotár et al., 2016). Handwriting exposes aspects of fine motor behaviour, but the relation between pen motion and diagnosis also depends on the task, device and participant characteristics (Impedovo and Pirlo, 2019).

We propose a new formulation of *pseudodynamic handwriting minimap signatures*: compact visual representations of task executions, derived from dynamic recordings. The term minimap follows work using compact source-code renderings as visual signatures (Gebara Jr. et al., 2025). Here, planar position supplies the image coordinates and colour carries measured pen attributes. A signature denotes this representation, not a person’s authentication signature. Starting from a monochrome trace, we add speed,

pressure, altitude and azimuth separately, then compare all four three-signal RGB combinations.

Handwriting classification has often used explicitly designed kinematic and pressure features. Drotár et al. compared such features with several classifiers on PaHaW (Drotár et al., 2016). Learned features provide another route, either from pen dynamics (Pereira et al., 2016) or from visual attributes of reconstructed writing (Moetesum et al., 2019). Dynamic information has also appeared in earlier handwriting images. Diaz et al. encode movement through acquired points and pen-up trajectories for PD detection (Diaz et al., 2019). Cilia et al. construct synthetic RGB images from handwriting dynamics for Alzheimer’s disease detection and compare them with binary images using transfer learning (Cilia et al., 2021). These precedents establish the broader idea of expressing dynamics visually.

Our formulation combines contact-only minimap rendering, within-recording percentile normalization and circular azimuth centring to map pen signals onto a fixed trajectory. We evaluate its components across all eight PaHaW tasks, using the same Local Phase Quantization (LPQ) descriptor, support vector machine (SVM) search and participant partitions for nine encodings. We then move encoding selection inside training and test controls that change where dynamic

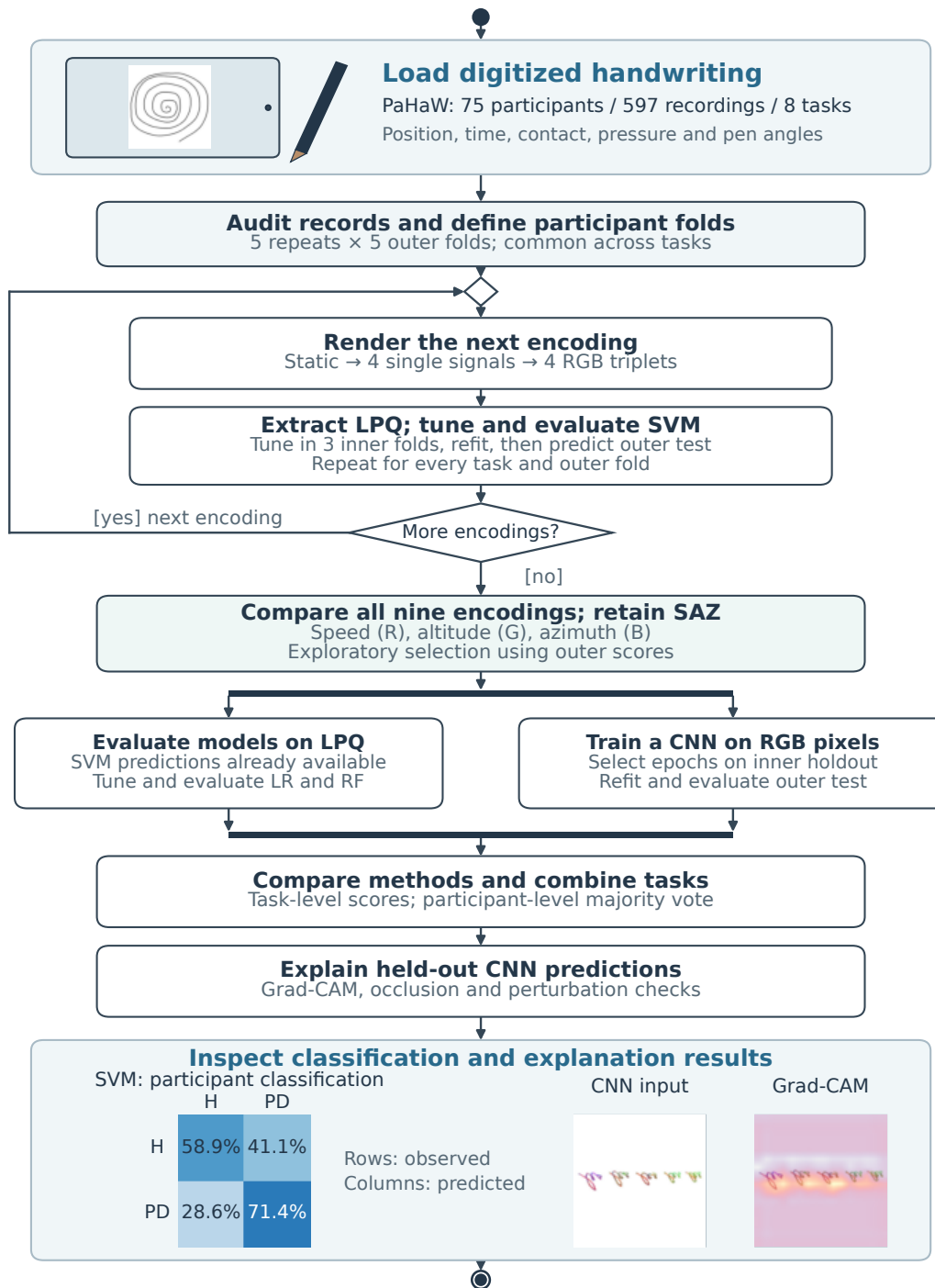


Figure 1: Illustrated activity diagram of the initial incremental study. Nine encodings are evaluated with LPQ and SVM using participant-separated outer folds. Selecting *SAZ* from the outer scores is exploratory. LR denotes logistic regression and RF random forest. XAI uses the scratch-trained CNN checkpoints responsible for held-out predictions. Later evaluations select encodings within training and add spatial controls and competitive references (Sections 4.4 and 7). The final illustrations show an aggregate SVM confusion matrix and one CNN explanation from the initial analyses. The tablet and stylus are schematic.

values occur along a stroke. All recordings and renderings from a participant remain in the same outer partition.

In the initial comparison, speed–altitude–azimuth raises observed macro F1 on four tasks, including the spiral, from 0.5915 to 0.6716. Its eight-task mean rises from 0.5643 to 0.5752. This ranking motivates the exploratory classifier and XAI analyses; it does not independently evaluate choosing the best encoding. When the encoding choice is evaluated inside training, the task mean is 0.5618, close to the re-evaluated static baseline of 0.5620. Spatial-control comparisons also leave the benefit of the original signal arrangement uncertain after correction.

Related studies do not all combine the same tasks. Casademunt González examines repeated *l*, *le* and *les*, whereas Drotár and Diaz use different larger subsets (Casademunt González, 2023; Drotár et al., 2016; Diaz et al., 2019). We fix the three repeated-pattern tasks before fitting competitive reference methods: a kinematic/pressure SVM and an ImageNet-pretrained ResNet18 with a trained linear head. Both share the participant partitions with our image-based SVMs. Keeping all eight tasks as a secondary comparison tests whether this shorter task set improves decisions for the same people.

On the three fixed tasks, training-selected LPQ+SVM has the highest observed participant-level score (0.6449), while kinematics has the highest task mean. The paired method differences remain uncertain. Task reduction helps the selected-encoding SVM numerically but lowers the static and transfer pipelines. The result concerns the interaction between task set and learning procedure, rather than a universally preferable subset.

The new selection procedures protect outer-test participants, but PaHaW has already informed the research choices. All stages therefore remain development-cohort evaluations. The earlier scratch-trained CNN provides a separate XAI case study: attribution concentrates on writing, yet its spatial agreement with occlusion is weak. Figure 1 shows the initial encoding loop and classifier branches; the later controlled evaluations retain those participant partitions.

## 2 BUILDING THE REPRESENTATION

To vary the information encoded in each signature, we first reconstruct the visible trace. We then map individual pen signals to colour and combine them without changing the trajectory or rendering geometry.

### 2.1 The Visible Trace

Let a recording contain samples

$$q_i = (x_i, y_i, t_i, s_i, z_i, a_i, p_i), \quad (1)$$

where  $s_i$  indicates surface contact,  $z_i$  is azimuth,  $a_i$  is altitude and  $p_i$  is pressure. The stored PaHaW columns place  $y$  before  $x$ ; the renderer restores their planar order.

The static image uses only position and the contact indicator. A segment between samples  $i$  and  $i + 1$  is drawn if both samples are on the surface:

$$S = \{i : s_i = 1 \text{ and } s_{i+1} = 1\}. \quad (2)$$

This preserves pen lifts as gaps instead of joining the end of one stroke to the beginning of the next. Neither the in-air trajectory nor the duration of a lift appears in the image. Contact points alone determine its bounding box, so an in-air excursion cannot change the scale of the visible writing.

The bounding box is centred on a  $512 \times 512$  white canvas with a 20-pixel margin, using one scale factor for both axes. Segments are drawn in RGB (30, 30, 30) with constant 4-pixel thickness. Rendering takes place at three times the target resolution and is reduced with Lanczos filtering. These settings remain fixed throughout the colour experiments.

The image retains form and stroke discontinuities, but fitting every record to its own canvas removes absolute size. Geometrically similar samples of different sizes can produce the same image, discarding information about absolute micrographia.

### 2.2 Adding One Signal

Four measurements are candidates for colour: speed ( $S$ ), pressure ( $P$ ), altitude ( $A$ ) and azimuth ( $Z$ ). Pressure is the tablet’s contact-pressure signal. Altitude describes pen inclination relative to the surface, while azimuth describes its orientation in the tablet plane. We first encode each signal alone, always in the red channel, with green and blue fixed at 30. This avoids changing the channel assignment while screening the signals.

For a surface segment, speed is calculated directly from the consecutive coordinates and timestamps:

$$v_i = \frac{\sqrt{(x_{i+1} - x_i)^2 + (y_{i+1} - y_i)^2}}{t_{i+1} - t_i}. \quad (3)$$

We use  $\log(1 + v_i)$  in the tablet’s native coordinate and time units. This compresses large values before the intensity mapping. Pressure and altitude use the arithmetic mean of the two endpoint measurements.



Figure 2: Building a pseudodynamic image from the same spiral trace. Rows show one control and one PD example, selected by a fixed identifier rule without reference to prediction quality. Columns move from static geometry to speed alone and two RGB triplets. The trajectory is unchanged; colour describes within-recording variation.

Azimuth requires a circular reference. We convert the recorded tenths of a degree to radians and compute the mean direction over contact samples. Each angle is centred on that direction and wrapped into  $[-\pi, \pi]$  before the endpoint average is taken. This reduces the effect of an arbitrary zero-degree boundary. It is a circular centring operation, not temporal unwrapping, and it does not remove every possible branch-cut artefact.

Each transformed signal is scaled within the recording being rendered. Let  $q_{.05}$  and  $q_{.95}$  be its fifth and ninety-fifth percentiles over surface segments. The normalized value is

$$u_i = \text{clip} \left( \frac{f_i - q_{.05}}{q_{.95} - q_{.05}}, 0, 1 \right), \quad (4)$$

with  $u_i = 0.5$  if the percentile range is degenerate. We map it to an integer channel value

$$c_i = \text{round}(30 + 190u_i). \quad (5)$$

The white background stays outside the semantic range. Antialiasing introduces intermediate intensities at boundaries, as in the static reconstruction.

Each record is normalized independently, without a cohort-wide reference distribution or class labels. Colour therefore describes relative variation within a recording: the same red value in two images need not represent the same absolute speed or pressure. Circular centring also removes absolute pen orientation. These quantities remain available to methods that use the physical measurements directly.

Table 1: Incremental colour encodings. A dash denotes a constant channel value of 30. *S*: speed; *P*: pressure; *A*: altitude; *Z*: azimuth.

| Encoding   | Red                                        | Green    | Blue     |
|------------|--------------------------------------------|----------|----------|
| Static     | –                                          | –        | –        |
| One signal | <i>S</i> , <i>P</i> , <i>A</i> or <i>Z</i> | –        | –        |
| <i>SPA</i> | <i>S</i>                                   | <i>P</i> | <i>A</i> |
| <i>SPZ</i> | <i>S</i>                                   | <i>P</i> | <i>Z</i> |
| <i>SAZ</i> | <i>S</i>                                   | <i>A</i> | <i>Z</i> |
| <i>PAZ</i> | <i>P</i>                                   | <i>A</i> | <i>Z</i> |

## 2.3 Combining Signals in RGB

We assign one scalar signal to each RGB channel and evaluate all four subsets of size three. Table 1 gives the assignments, and Figure 2 illustrates the progression. All combinations use the same geometry, background and rendering settings as the static and isolated-signal images.

The comparison holds the classifier procedure fixed. Because its descriptor concatenates the same histogram construction for each channel, permuting the RGB channels only permutes three feature blocks. Standardization followed by a linear or radial-basis SVM is invariant to that common permutation, apart from numerical effects. The RGB assignments thus specify a reproducible representation without requiring six permutations of every triplet for the SVM experiment. This argument does not extend to every image model; the later CNN receives one fixed RGB assignment.

Segments are drawn in acquisition order. A later segment can overwrite an earlier colour at a crossing,

Table 2: Task inventory. T1 contains 36 PD and 36 H records; each other task contains 37 PD and 38 H records.

| ID | Task                             | Records |
|----|----------------------------------|---------|
| T1 | Archimedean spiral               | 72      |
| T2 | Repeated <i>l</i>                | 75      |
| T3 | Repeated <i>le</i>               | 75      |
| T4 | Repeated <i>les</i>              | 75      |
| T5 | Repeated <i>lektorka</i>         | 75      |
| T6 | Repeated <i>porovnat</i>         | 75      |
| T7 | Repeated <i>nepopadnout</i>      | 75      |
| T8 | <i>Tramvaj dnes už nepojede.</i> | 75      |

and the image has no separate layer for repeated visits to the same position. It preserves spatially located signal variation, not the complete sample sequence. Calling the result pseudodynamic reflects that distinction.

### 3 DATA AND EVALUATION

Each recording produces nine alternative images, all derived from the same participant. We compare encodings on common participant partitions and define separate endpoints for individual tasks and decisions combined across tasks. The initial descriptive ranking, the training-selected encoding evaluation and the competitive task-subset comparison are distinct stages with explicitly stated selection rules.

#### 3.1 Participants and Tasks

PaHaW provides eight tasks recorded with a Wacom Intuos 4M tablet (Drotár et al., 2016). We calculate speed from the recorded timestamps rather than an assumed sampling interval. Our local inventory contains 597 recordings from 75 participants: 37 with PD and 38 controls. Table 2 specifies the task set. Three spiral recordings are absent from the local corpus; the other tasks have one recording for every participant. We retain all available records and do not interpret missing files as quality exclusions.

An input audit checked file counts, sample counts, finite values and timestamp increments. Five files have a declared sample count different from the number of rows present. The renderer uses the rows actually available. There are 253 negative timestamp increments in 151 recordings, but all 1,065,788 segments with contact at both endpoints have positive increments. Equation 3 therefore requires no interpolation or imputation for the rendered contact segments. The later kinematic baseline also uses in-air intervals and applies the timestamp rules in Section 7. None of the signal normalization ranges was degenerate.

The participant is the independent observational unit; tasks and alternative renderings do not enlarge the cohort. Mean age is 69.32 years in PD and 62.42 in H. Age and clinical fields are excluded from classifier inputs, but handwriting may still carry this confounding.

#### 3.2 A Common Partition for Every Task

We construct stratified folds from one row per participant, then propagate the assignments to every task and rendering (Figure 3). Five repetitions of five outer folds produce a test prediction for every available record in each repetition. All tasks from a participant are in the same outer fold. The spiral task uses the participants for whom a record exists.

Within each outer training partition, three participant-separated inner folds choose the classical model hyperparameters. Feature standardization is fitted on the corresponding training portion, never on its validation or test participants. The selected model is refitted on all outer training participants and evaluated once on the outer test set. The rendering is independent per record, so it has no parameters learned from the other records before splitting.

In the initial eight-task evaluation, each fixed encoding and classifier yields 200 outer models and 2,985 outer-test predictions. These are repeated predictions of 597 records from 75 participants. Every later stage reuses the saved outer assignments. The follow-up stages use three inner folds common across tasks, intersected with available recordings.

The initial highest-ranked encoding was selected from outer scores for the first classifier comparison and XAI analysis. That stage protects hyperparameter fitting but not representation choice. Sections 4.4 and 7 subsequently evaluate encoding selection within training. They estimate different learning procedures and retain the limitation that earlier exploration used the same cohort.

#### 3.3 Metrics and Paired Comparisons

Macro F1 gives equal weight to the two diagnosis classes:

$$F_{\text{macro}} = \frac{F_{1,H} + F_{1,PD}}{2}. \quad (6)$$

For each task and repetition, we concatenate the five outer-test folds and compute the metric once. We then average the five resulting scores. For the initial representation and classifier rankings, the endpoint is the unweighted mean across all eight tasks:

$$\bar{F}_{\text{tasks}} = \frac{1}{8 \cdot 5} \sum_{t=1}^8 \sum_{r=1}^5 F_{t,r}. \quad (7)$$

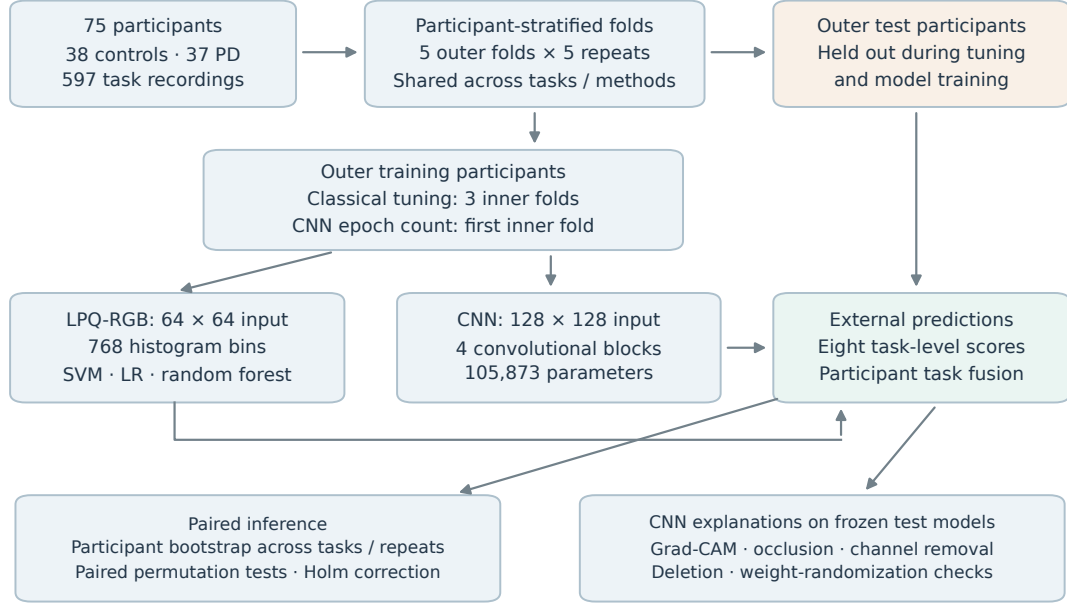


Figure 3: Participant separation in the initial model comparisons. The scratch-trained CNN selects its epoch count on the first inner holdout; the classical models average three inner-fold scores. Outer assignments are common across tasks and methods. This diagram describes fitting for a fixed encoding; the initial outer-score representation choice is exploratory. Follow-up encoding selection uses training-only folds as described in Sections 4.4 and 7.

We also retain accuracy, balanced accuracy, PD sensitivity, H specificity, positive-class F1 and ROC AUC. Section 6 defines a separate participant-level prediction by combining tasks.

We calculate 95% percentile confidence intervals from 10,000 diagnosis-stratified bootstrap draws of participants. A draw preserves all available tasks and repeated predictions for each selected person. For a paired contrast, the same draw is applied to both methods. The randomization test exchanges method predictions per participant, jointly across tasks and repetitions, using 10,000 participant-wise swap patterns. It tests the difference between fixed sets of predictions, not a shuffled-label learning pipeline.

For the initial analyses, Holm adjustment covers 36 encoding pairs; two separate 32-contrast task families for isolated signals and triplets against static; six classifier pairs at each endpoint; three SVM fusion-rule pairs; and eight task-specific CNN-SVM contrasts. Each follow-up section defines its additional families. Correction is within each declared family, not across all decisions in the broader research history. The intervals are pointwise, not simultaneous.

Both resampling procedures leave models fixed. Intervals and tests describe conditional uncertainty in these predictions, without the full variability of re-training or selection.

## 4 FROM SHAPE TO COLOUR

The common participant partitions allow paired comparisons from static traces to isolated signals and RGB combinations. We hold LPQ extraction and SVM selection fixed so that each comparison changes the image encoding.

### 4.1 A Fixed Descriptor for the Comparison

Local Phase Quantization describes texture through local Fourier phase information (Ojansivu and Heikkilä, 2008). We apply the same implementation independently to each RGB channel.

Before extraction, foreground is detected by any channel intensity below 250. We crop its bounding box, add a margin equal to 5% of the longer dimension, pad to a square with white, and resize to  $64 \times 64$  by bilinear interpolation. LPQ filters each channel with four complex  $5 \times 5$  kernels at frequency offsets  $(1,0)$ ,  $(0,1)$ ,  $(1,1)$  and  $(1,-1)$ , scaled by  $1/5$ . The signs of their real and imaginary responses form an eight-bit code at each pixel, using reflected boundaries. We normalize a 256-bin histogram by its count, take its element-wise square root, and concatenate the channels. The 768-attribute vector includes the padded background. No decorrelation or spatial pyra-

mid is applied.

The static RGB channels are identical, so the static vector repeats the same histogram three times. This keeps the interface and nominal dimension constant; it does not create additional static information. For the isolated signals, two channels remain constant along the trace and retain its geometric pattern.

Standardization and SVM fitting occur inside each training partition. The search contains five linear models with  $C \in \{0.01, 0.1, 1, 10, 100\}$  and sixteen radial-basis models with  $C \in \{0.1, 1, 10, 100\}$  and  $\gamma \in \{\text{scale}, 10^{-4}, 10^{-3}, 10^{-2}\}$ . We use inverse-frequency class weights. The candidate with the highest mean inner-fold macro F1 is chosen, with ties resolved by the stored candidate order. The SVM in this encoding comparison learns from LPQ image texture; it receives no original signal measurements as tabular features (Cortes and Vapnik, 1995).

## 4.2 The Static Baseline

The static reconstruction reaches a task-mean macro F1 of 0.5643. Its best task is repeated *les* (T4), with 0.6502 and a participant-bootstrap interval of [0.5582, 0.7351]. The spiral reaches 0.5915 and T5 reaches 0.5038. All eight tasks remain in the subsequent encoding comparison.

## 4.3 Isolated Signals and RGB Combinations

Figure 4 orders the encodings by Equation 7. Altitude gives the highest task mean among the isolated signals at 0.5718, followed by azimuth at 0.5688. Speed alone reaches 0.5625 and pressure 0.5568, both below the static mean of 0.5643.

The individual-task contrasts vary in direction. For example, speed changes the spiral score from 0.5915 to 0.6153 but changes repeated *les* from 0.6502 to 0.5994. Altitude increases the spiral score to 0.6294 and changes T4 only slightly, to 0.6548. None of the 32 isolated-signal versus static contrasts survives its Holm correction.

Combining signals produces a similarly narrow ranking. Starting from *PAZ*, replacing pressure, azimuth or altitude with speed gives *SAZ*, *SPA* or *SPZ*, respectively. *SAZ* leads at 0.5752, only 0.0006 above *PAZ*; *SPA* reaches 0.5697 and *SPZ* reaches 0.5574. These results provide no basis for declaring pressure intrinsically uninformative. LPQ summarizes normalized image patterns, not the physical measurements directly.

The paired *SAZ*–static difference is +0.0109 with a 95% interval of [−0.0146, 0.0370]. Its random-

ization  $p$  is 0.4096 and Holm-adjusted  $p$  is 1.0000 across the 36 encoding pairs. The *SAZ*–*PAZ* interval is [−0.0185, 0.0198], also with adjusted  $p = 1.0000$ . We carry *SAZ* forward as the numerical leader in this exploratory comparison.

Its per-task changes explain the smaller average gain: the spiral improves by 0.0802, T6 by 0.0524, and T7 by 0.0455, whereas T2 falls by 0.0612. None of its eight task-specific contrasts against static remains significant within the 32-comparison triplet family.

## 4.4 Evaluating the Encoding Choice Within Training

The ranking above evaluates fixed encodings. We next evaluate the procedure that chooses an encoding, retaining the outer participant partitions while introducing three inner folds common across tasks. Within each outer training set, each task and encoding receives the same 21-candidate SVM search. We choose one global encoding by the unweighted mean of its eight selected task scores. Ties follow the declared order, with static first when eligible. We evaluate a fixed-static policy, selection among all nine encodings, and selection restricted to the eight dynamic encodings.

For participant fusion in this stage, a further internal layer compares the three rules from Section 6. Each of three intermediate training sets repeats encoding and SVM selection in its own three folds before predicting the intermediate validation participants. These predictions choose the fusion rule; the encoding and task models are then selected anew on the full outer training set. Sigmoid calibration uses training-only folds with a single final estimator (`ensemble=False`); reliability remains  $\max(F_{\text{inner}} - 0.5, 0.01)$ . All choices are saved before outer-test evaluation.

The re-evaluated static task mean is 0.5620. Selection among nine encodings gives 0.5618, and dynamic-only selection gives 0.5643. Their paired differences from static are −0.0002 [−0.0192, 0.0190] and 0.0023 [−0.0172, 0.0221]. These are results of adaptive procedures, not new estimates for a fixed *SAZ* encoding. The altered inner folds and calibration also prevent attributing every difference from the initial stage solely to correcting encoding selection.

The chosen encoding varies substantially. In the dynamic-only policy, azimuth is selected in five of the 25 outer training sets and *SAZ* in four; no encoding predominates. All-nine selection chooses static once. These overlapping training sets describe selection sta-

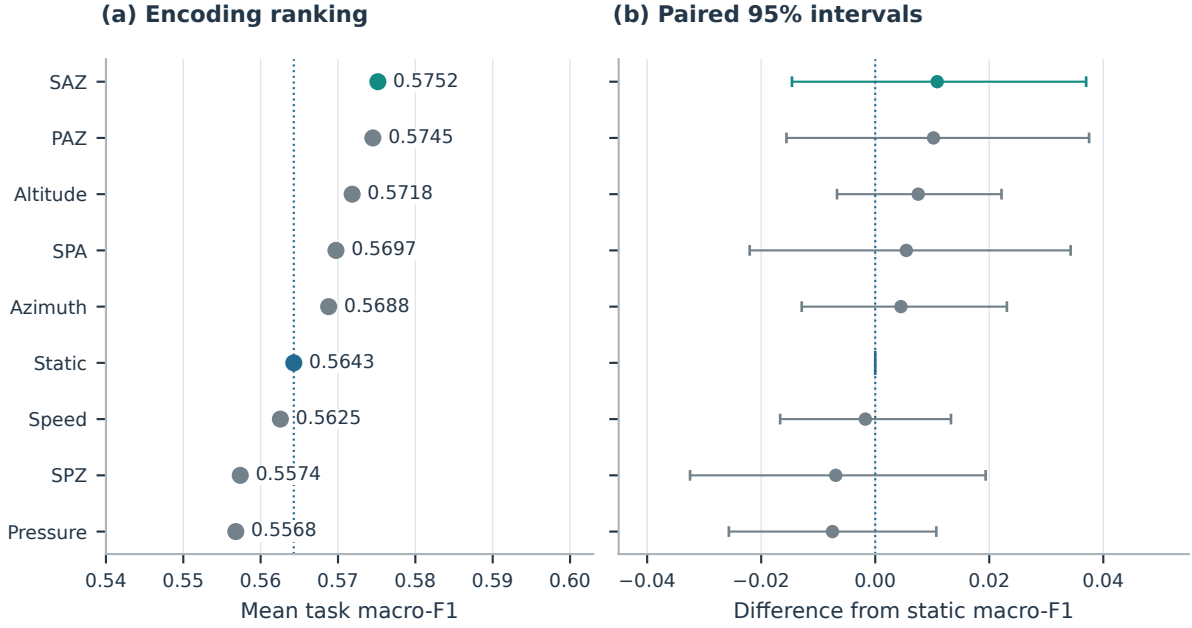


Figure 4: Encoding comparison with SVM. Left: unweighted mean of task-specific macro F1 over eight tasks and five repetitions. Right: paired differences from static with conditional 95% participant-bootstrap intervals. The complete family of 36 pairwise comparisons supports no difference after Holm adjustment.

bility, not 25 independent cohorts. Internally selected fusion yields macro F1 0.6171 for static, 0.5991 for all-nine selection and 0.6017 for dynamic-only selection. Neither contrast against static survives its two-comparison Holm family.

#### 4.5 Controls of the Signal–Position Association

Adding colour changes both signal content and image texture. We test its spatial association using two controls: a nonzero circular shift by 25–75% of the segment count within each pen-down stroke, and an unrestricted joint permutation within that stroke. Channels move together, preserving their joint vectors and cross-channel relationships. Single-segment strokes remain unchanged. No values move across pen lifts.

Both controls retain geometry, line width, normalization and the multiset of vectors within each stroke. They need not preserve the raster colour histogram: segments have unequal lengths, overlap and undergo antialiasing. Joint permutation additionally disrupts local continuity, so a performance difference would not isolate physiological signal placement from induced texture changes.

Each control transforms training and test records using three fixed seeds. For each seed, its own eight transformed encodings receive the same internal SVM and encoding search as the authentic dynamic-

only policy. We average metrics over seeds rather than selecting a seed or ensembling their predictions. Participant bootstrap draws and swaps retain every person’s tasks, repetitions and seeds together. This control experiment evaluates task scores only.

Circular shifts yield task-mean macro F1 0.5561 and permutations 0.5442, compared with 0.5643 for authentic dynamic-only selection (Figure 6). The authentic-minus-permutation difference is 0.0201 [0.0010, 0.0399], with unadjusted  $p = 0.0462$  and Holm-adjusted  $p = 0.1848$ . A pointwise interval excluding zero is compatible with an adjusted test that does not reject. The circular-shift contrast is 0.0082 [−0.0102, 0.0264].

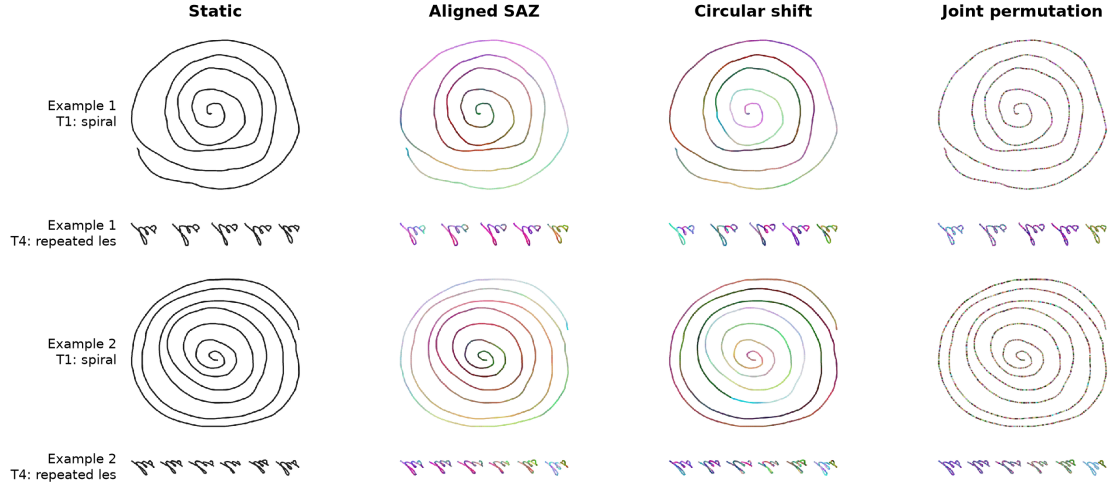
The four global task contrasts comprise both authentic policies against static and dynamic-only selection against each control. None survives Holm adjustment. The corresponding 32 task-specific contrasts also remain unsupported in a separate exploratory family. These results neither establish a spatial-dynamic advantage nor demonstrate equivalence with static images.

## 5 FROM TEXTURE TO LEARNED FEATURES

We return to the initial exploratory ranking to examine how its numerically leading representation inter-

### Spatial signal controls on the same handwriting

SAZ channels: red = speed; green = altitude; blue = azimuth



Illustrative SAZ examples, with the fixed seed 20260909. The first two participants of the previously fixed, diagnosis-independent example selection contribute T1 and T4. Geometry, pen-down segments and stroke width are unchanged. Circular shifts preserve cyclic vector order within each stroke; joint permutations do not. Both retain the joint-signal multiset of each stroke. The four views of a recording share one display crop; raster pixel histograms need not be identical. These examples are not selected by classifier performance.

Figure 5: Within-stroke controls illustrated with SAZ. The first two participants in a fixed identifier-based example order contribute T1 and T4; the seed is 20260909. Each row shares geometry, line width and display crop. Circular shifts and joint permutations change signal assignment within each pen-down stroke while preserving its multiset of joint signal vectors. The examples were not selected by classifier performance.

acts with the classifier. Keeping *SAZ* and the participant partitions fixed, we compare models fitted to LPQ features with a CNN trained from scratch. These results are conditional on the earlier outer-score selection of *SAZ*; they are distinct from the adaptive policies evaluated above and the competitive comparison that follows.

## 5.1 Linear and Tree-Based Alternatives

With the *SAZ* image fixed, SVM, regularized logistic regression and Random Forest use the same 768 LPQ attributes. Logistic regression provides a linear baseline. We search  $L_1$  and  $L_2$  penalties with  $C \in \{0.001, 0.01, 0.1, 1, 10, 100\}$ , using the liblinear solver and inverse-frequency class weights. Standardization remains inside the training pipeline.

Random Forest supplies a different form of non-linear modelling through an ensemble of decision trees (Breiman, 2001). Each candidate uses 100 trees, the Gini criterion and class weights calculated within bootstrap samples. The search combines maximum-feature choices  $\{\sqrt{d}, 0.25d\}$  with minimum leaf sizes  $\{1, 2, 4\}$ , where  $d = 768$ . Tree depth is unrestricted. Both algorithms reuse the same three-fold inner search and the same outer test assignments as SVM.

## 5.2 Learning Directly from Pixels

We use a compact CNN trained from scratch as a direct pixel-learning baseline, following the use of convolutional models in document recognition (LeCun et al., 1998). Its architecture is fixed throughout the comparison.

The input follows the same foreground crop and square padding as LPQ but is resized to  $128 \times 128$ . Pixel values are mapped from  $[0, 255]$  to  $[-1, 1]$  with fixed constants. Four blocks contain a bias-free  $3 \times 3$  convolution, batch normalization, ReLU and  $2 \times 2$  max pooling. Their channel counts are 16, 32, 64 and 128. A dropout layer with probability 0.3 and one linear output act on the flattened  $128 \times 8 \times 8$  feature map. The network has 105,873 trainable parameters.

Training uses AdamW with learning rate  $10^{-3}$ , weight decay  $10^{-4}$ , batch size 16 and binary cross-entropy with positive weight equal to the H/PD training-count ratio. Training augmentation uses rotations up to  $3^\circ$ , translations up to 3% and scale changes between 0.95 and 1.05. White fills exposed pixels; writing is never flipped.

For each outer fold and task, the first of the saved inner splits chooses the number of epochs. Selection uses validation macro F1, with binary cross-entropy as the tie-breaker. Training runs for at most

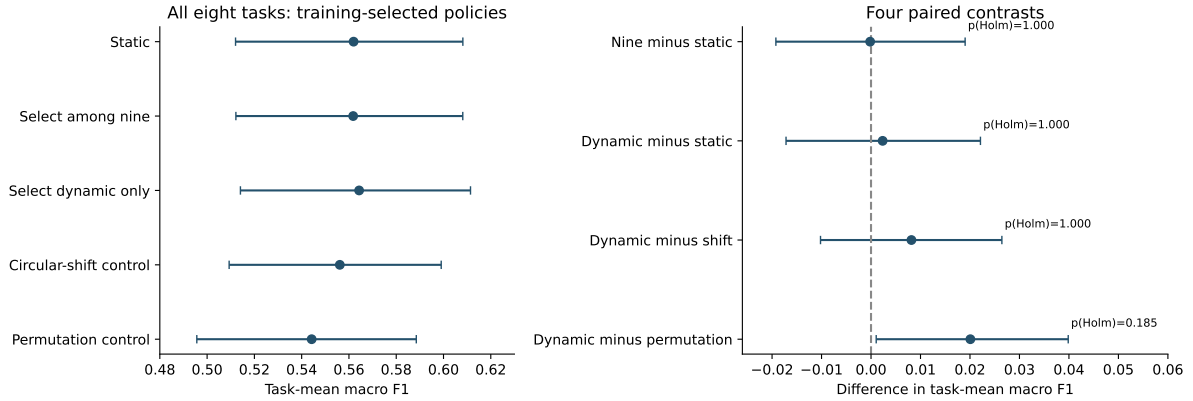


Figure 6: Training-selected encoding and spatial controls on all eight tasks. Left: task-mean macro F1. Right: paired effects with pointwise 95% participant-bootstrap intervals. Control points average three fixed seeds; they do not treat seeds as independent participants. The four contrasts form one Holm family, with no supported difference.

60 epochs, with a patience of ten and stopping enabled after epoch twelve. The selected epoch itself may be earlier than twelve. We then reinitialize the network and train on all outer training participants for exactly that number of epochs. The outer test set is used only for the saved evaluation. Unlike the classical grid searches, the CNN epoch selection uses one inner holdout rather than an average over all three inner folds.

Selected counts range from 1 to 48 epochs, with median 15. All 200 outer checkpoints are retained for XAI.

### 5.3 Task-Level Comparison

SVM leads the task-mean ranking at 0.5752, followed by logistic regression at 0.5674 and Random Forest at 0.5599. Their differences are small relative to the paired uncertainty. SVM minus logistic regression is 0.0077, with interval  $[-0.0074, 0.0230]$  and Holm-adjusted  $p = 0.6563$ . SVM minus Random Forest is 0.0152, with interval  $[-0.0066, 0.0365]$  and adjusted  $p = 0.5645$ .

The CNN reaches 0.5105. The SVM–CNN contrast is 0.0646, with interval  $[0.0277, 0.1021]$  and adjusted  $p = 0.0054$  in the six-pair classifier family. Random Forest and logistic regression also exceed the CNN, with adjusted  $p = 0.0460$  and  $p = 0.0255$ , respectively. The inference follows the fixed-prediction procedure in Section 3.

The task pattern in Figure 7 differs from the overall ranking. Random Forest is strongest on T4 at 0.6791, even though its eight-task mean is below SVM. The CNN is slightly above SVM on T2 and T5, two tasks on which SVM itself performs poorly. Its task-specific differences against SVM do not sur-

vive the family of eight contrasts.

LPQ-based models perform better under these choices, which differ in more than architecture: LPQ uses  $64 \times 64$  inputs, the CNN uses  $128 \times 128$ , and their selection budgets differ. The comparison does not isolate the effect of architecture or evaluate the pretrained features used in earlier enhanced-image work (Diaz et al., 2019).

## 6 COMBINING HANDWRITING TASKS

For the initial fixed-SAZ models, late fusion combines available task predictions into one decision per participant. Because all tasks share the same participant folds, every model contributing to a held-out person’s decision was fitted without that person.

We first considered three SVM rules over every available task: the mean calibrated PD probability, a reliability-weighted mean, and majority voting. Sigmoid calibration uses the three inner participant folds with ensemble averaging. Reliability is derived from inner-validation macro F1, with weight  $\max(F_{\text{inner}} - 0.5, 0.01)$ . We do not select a subset of tasks according to outer-test performance. A missing spiral is omitted from that participant’s denominator.

The probability mean, specified before the fusion-rule comparison, reaches macro F1 0.6341, with interval  $[0.5417, 0.7189]$ . Reliability weighting reaches 0.6447 and majority voting 0.6488. Majority minus probability mean is 0.0147, with interval  $[-0.0317, 0.0618]$  and adjusted  $p = 1.0000$ ; none of the three rule comparisons survives correction.

We use majority voting as a common descriptive participant endpoint for all four classifiers. If  $T_j$  is

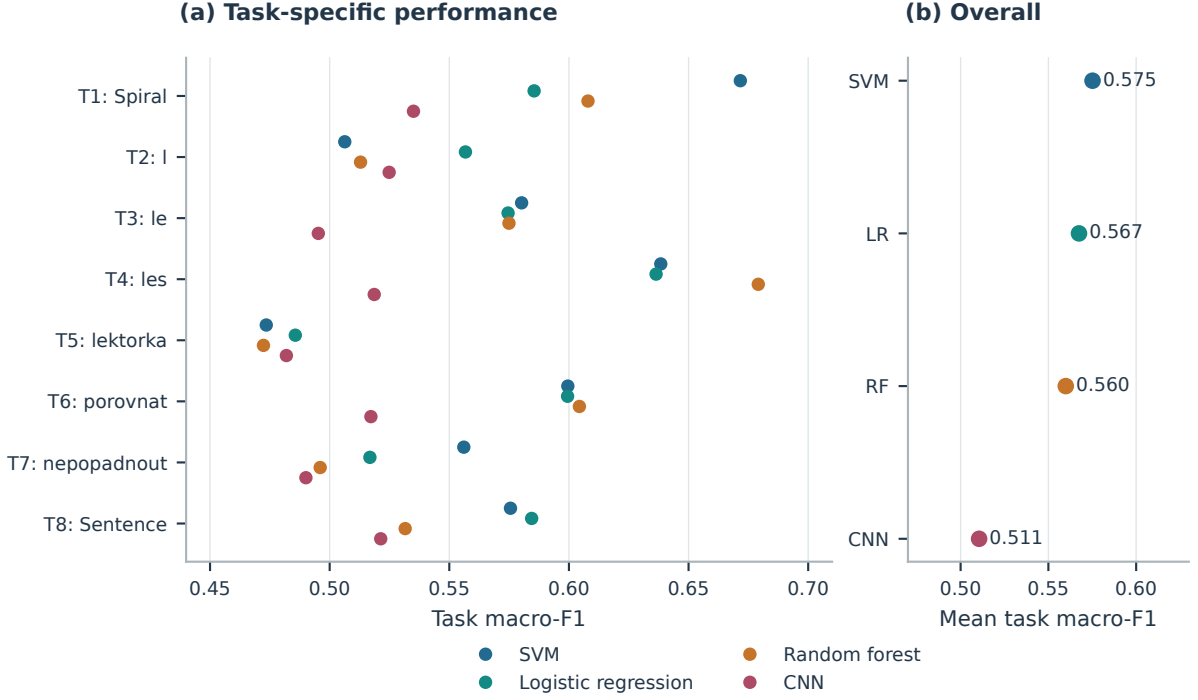


Figure 7: Classifier comparison on SAZ. Left: the task-specific results, each averaged over five repetitions. Right: the unweighted task mean. The classical methods use LPQ; the CNN uses pixels. Inferential contrasts concern the fixed predictions and this selected representation.

the set of tasks available for participant  $j$ , its score in repetition  $r$  is

$$s_{j,r} = \frac{1}{|\mathcal{T}_j|} \sum_{t \in \mathcal{T}_j} \hat{y}_{j,t,r}, \quad \hat{y}_{j,r} = \mathbf{1}[s_{j,r} \geq 0.5]. \quad (8)$$

Tied votes are assigned to PD by this fixed threshold. We compute the participant-level metric separately for each repetition and average the five scores. We do not average probabilities across repetitions to create another ensemble.

Figure 8 places logistic regression at 0.6498 and SVM at 0.6488 after fusion, both with mean accuracy 0.6507. Random Forest reaches macro F1 0.6259 and the CNN 0.5544. Logistic regression’s macro F1 exceeds SVM by only 0.0010, with interval  $[-0.0398, 0.0403]$  and adjusted  $p = 1.0000$ . No pairwise fusion contrast remains significant after Holm correction. Failure to detect a difference is not a formal equivalence result.

The fused SVM has PD sensitivity 0.7135 and H specificity 0.5895. Logistic regression has sensitivity 0.7081 and specificity 0.5947. Their error rates are not symmetric despite near-balanced class counts. The PD decision on tied votes contributes to the interpretation of this operating point; no alternative threshold was chosen from the outer labels. The vote fraction gives an SVM AUC of 0.6808 and a logistic-

regression AUC of 0.6658, even though logistic regression ranks slightly higher by thresholded macro F1.

These participant-level scores and the earlier task means summarize different predictions; their numerical difference is not tested as a paired improvement.

## 7 COMPETITIVE BASELINES ON A FIXED TASK SUBSET

The preceding comparisons examine image texture and a CNN trained from scratch. We now broaden the reference methods to include the original movement signals and pretrained visual features. A task set drawn from related work allows us to examine these alternatives under a shorter handwriting protocol, while retaining all eight tasks as a secondary comparison.

### 7.1 Task Set and Evaluation Protocol

The primary set comprises repeated *l*, *le* and *les* (T2–T4), the tasks reported in the reference thesis (Casademunt González, 2023). We fix this set before fitting the new models. Combining its three tasks by vot-

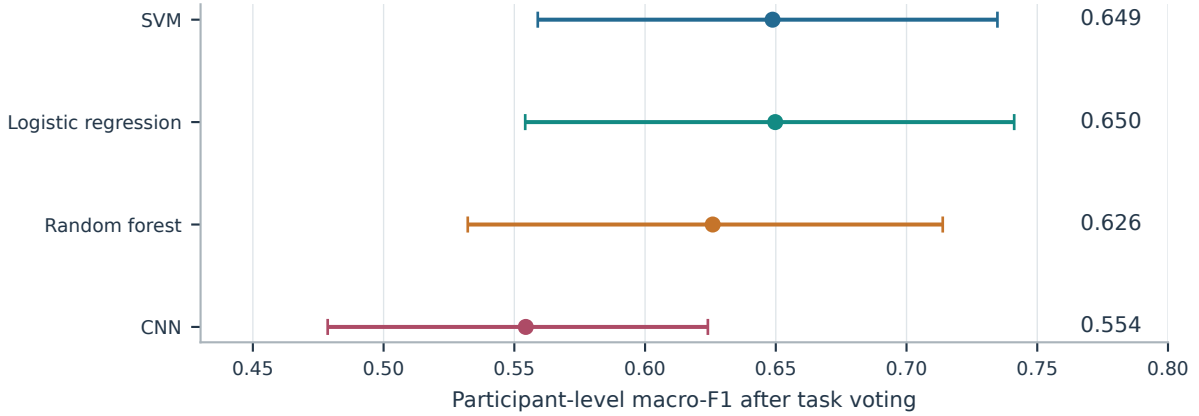


Figure 8: Initial fixed-SAZ classifier comparison: one prediction per participant from voting over available tasks. Points average macro F1 over five repetitions; intervals are conditional 95% participant-bootstrap intervals. These scores have a different prediction unit from task means. None of the six paired fusion comparisons remains significant after Holm adjustment.

ing is our adaptation; the thesis does not report this ensemble. We evaluate four methods: static LPQ with SVM, LPQ with internal selection among nine encodings, a kinematic/pressure SVM, and a frozen ResNet18 with a trained linear classification head. All eight tasks form a separately specified secondary evaluation.

The methods share the original 25 outer participant partitions and three common inner participant folds. For each task and eligible encoding, mean inner-fold macro F1 selects the classifier parameters. The two image methods that allow encoding selection choose one encoding for the task set by averaging these selected inner scores across its tasks. Candidate order resolves ties, with static first among encodings. Each method makes its own choices using only outer-training participants. Feature standardization is fitted within the corresponding training partition; selected models are then refitted on the complete outer-training set. The three SVM methods retain the same 21-candidate linear/RBF search.

Our primary endpoint is participant-level macro F1 from fixed majority voting over T2–T4. Voting uses all available predicted task labels, assigns ties to PD and has no calibration, learned weights or rule selection. Missing spirals are omitted from eight-task votes. We pool outer folds before scoring each repetition and average the five repetition scores. Task-mean macro F1 is secondary: means over three and eight tasks describe different task populations. Their values alone cannot establish a benefit from reducing the task set. The paired fusion comparison instead evaluates the same 75 participants, although changing the task set can also change the internally selected encoding.

Participant bootstrap intervals and paired predic-

Table 3: Fixed kinematic/pressure descriptor. Each distribution contributes mean, population standard deviation, median, 5th and 95th percentiles, and interquartile range.

| Family                                                                                  | Count |
|-----------------------------------------------------------------------------------------|-------|
| Acquisition times, contact fraction, stroke counts, path lengths and contact dimensions | 11    |
| Contact/air stroke duration and length distributions                                    | 24    |
| Contact/air speed, acceleration and jerk distributions                                  | 36    |
| Contact pressure and absolute pressure-rate distributions                               | 12    |
| Total                                                                                   | 83    |

tion swaps follow Section 3. Holm adjustment covers the six method pairs separately at each endpoint; the four within-method contrasts between three-task and eight-task fusion form another family. These estimates remain conditional on saved predictions from a cohort already used during development.

## 7.2 Kinematics and Pressure

The signal baseline retains quantities removed by image fitting and per-record colour normalization. Its 83 fixed attributes cover the families in Table 3, motivated by earlier kinematic and pressure measurements on PaHaW (Drotár et al., 2016). This is an independently specified descriptor, with no supervised feature screening or claim to reproduce their exact pipeline.

Coordinates and pressure remain in native tablet units; timestamp differences are divided by 1,000 to obtain seconds. Velocity vectors are secants at interval midpoints. Successive vector differences over midpoint time differences yield acceleration and jerk, whose magnitudes are summarized. Derivatives never

cross pen lifts or invalid intervals. No smoothing is applied, so quantization can influence higher derivatives.

A pre-fit audit excluded 255 of 1,364,022 intervals: 253 negative increments and two positive jumps exceeding 60s. None occurred within contact. Positive intervals up to 60s are retained, including 145 gaps of 1–26.61s; their velocities describe average movement across the gap. Acquisition time sums valid intervals, dividing each state-transition interval equally between contact and air. Stroke lengths and derivatives exclude transitions. Six zeros represent an absent distribution, recorded separately in diagnostics: 56 recordings lack in-air distributions and 57 lack an in-air jerk distribution. These conventions preserve every recording while making incomplete temporal information explicit.

### 7.3 Transfer from a Pretrained CNN

ResNet18 (He et al., 2016) supplies 512 features from its global average-pooling layer using the fixed torchvision IMAGENET1K\_V1 weights. Images undergo the same foreground crop and square white padding as LPQ, followed by bilinear resizing to  $224 \times 224$  and fixed ImageNet channel normalization. The backbone remains in evaluation mode, with convolutional weights and batch-normalization statistics frozen. There is no augmentation or end-to-end fine-tuning.

A class-balanced,  $L_2$ -regularized logistic head is fitted to standardized features. Its inner search evaluates seven values,  $C \in \{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1, 10, 100\}$ , and selects among the same nine image encodings as LPQ. This comparison therefore concerns complete learning procedures with different feature families and search budgets. It evaluates fixed-feature CNN transfer on PaHaW, without attributing any difference solely to architecture.

### 7.4 Comparison on the Three Fixed Tasks

Training-selected LPQ+SVM has the highest observed participant macro F1, 0.6449 (Table 4). Its difference from static LPQ is 0.0307, with interval  $[-0.0107, 0.0748]$ , unadjusted  $p = 0.1815$  and adjusted  $p = 1.0000$ . Its differences from the kinematic and transfer pipelines are also uncertain. All six primary adjusted  $p$  values equal one; this does not establish equivalence among the methods.

The selected-encoding SVM has PD sensitivity 0.5892 and H specificity 0.7053. Thus its macro F1

does not imply equally strong performance in both classes. Its vote-fraction AUC is 0.6139, whereas the kinematic model reaches 0.6691 despite a lower thresholded macro F1. Macro F1 evaluates the fixed voting decision; AUC evaluates ranking by the coarse vote fractions.

Kinematics have the highest individual-task mean, 0.6074, although LPQ-selected fusion is numerically higher. An average task score does not encode the coincidence of errors across tasks. We report the two endpoints separately and do not test their numerical difference as an improvement. Neither the six task-mean contrasts nor the primary fusion contrasts supports a method advantage after correction.

### 7.5 Does Reducing the Task Set Help?

With all eight tasks, fixed majority fusion reaches macro F1 0.6330 for static LPQ, 0.6166 for selected LPQ, 0.5941 for kinematics and 0.6225 for transfer features (Figure 9). These values use a fixed vote rather than the internally selected fusion rule of Section 4.4; their differences across sections do not represent contradictory estimates of the same procedure.

Moving to three tasks changes participant macro F1 by  $-0.0188$  for static LPQ,  $+0.0283$  for selected LPQ,  $+0.0338$  for kinematics and  $-0.0204$  for transfer features. Every interval includes zero and every adjusted  $p$  equals one in the four-comparison family (Figure 10). For selected LPQ, the interval is  $[-0.0358, 0.0950]$ . The shorter set is therefore a plausible option for further study, with a numerical gain for two pipelines and losses for two others; its general advantage is unproven.

The transfer method has the highest eight-task mean at 0.5813, compared with 0.5618 for training-selected LPQ. Their difference, 0.0195  $[-0.0152, 0.0543]$ , is not supported after correction. This result limits the earlier conclusion about the small scratch-trained CNN: it does not justify a general ranking of texture descriptors above pretrained CNN features.

## 8 WHAT DOES THE SCRATCH-TRAINED CNN USE?

We return to the small CNN trained from scratch in Section 5. Its saved checkpoints support a separate analysis of how that model uses the encoded images; these explanations do not describe the ResNet18 transfer pipeline. We examine whether its predictions depend on the handwriting region or on image placement and background, then compare spatial explanations and responses to perturbations.

Table 4: New competitive comparison on T2/T3/T4, fixed before fitting. Task mean and participant fusion are distinct endpoints. Intervals are conditional 95% participant-bootstrap intervals for fusion macro F1. Each number averages five repetitions after pooling outer folds. None of the six primary method contrasts survives Holm correction.

| Method                          | Task macro F1 | Fusion macro F1 | Fusion 95% CI    | Fusion acc. (%) |
|---------------------------------|---------------|-----------------|------------------|-----------------|
| Static LPQ + SVM                | 0.5815        | 0.6143          | [0.5194, 0.7049] | 61.60           |
| Training-selected LPQ + SVM     | 0.5773        | 0.6449          | [0.5502, 0.7340] | 64.80           |
| Kinematic/pressure SVM          | 0.6074        | 0.6279          | [0.5409, 0.7106] | 62.93           |
| ResNet18 transfer + linear head | 0.5710        | 0.6021          | [0.5136, 0.6873] | 60.27           |

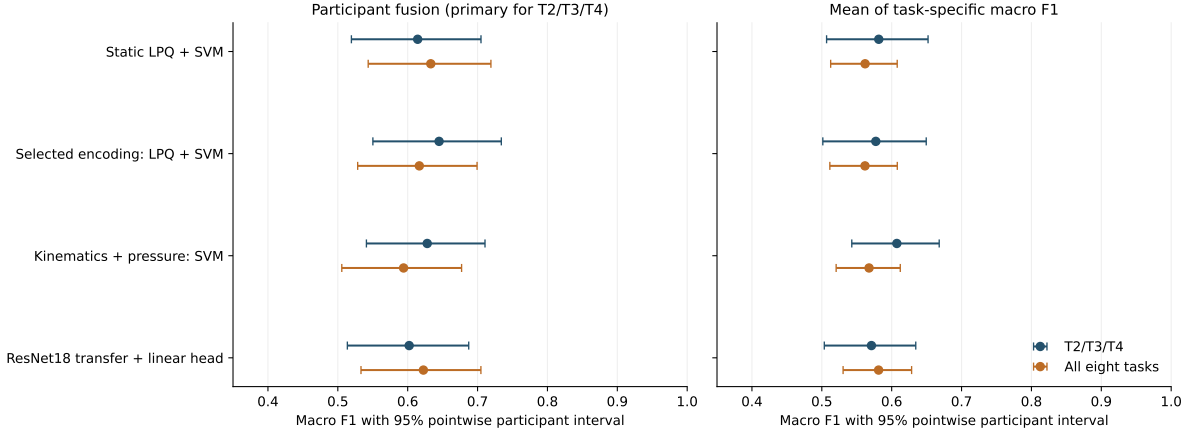


Figure 9: Four learning procedures on the three literature-defined tasks and on all eight tasks. The left panel compares participant decisions; the right panel compares task means, which refer to different task populations. Intervals are pointwise 95% participant-bootstrap intervals. Each method selects its parameters and, when eligible, its encoding within training. The primary six-pair family is the T2/T3/T4 participant comparison.

## 8.1 Explanations of Held-Out Predictions

We apply XAI after freezing all outer checkpoints. Each image is explained by the exact checkpoint that produced its held-out prediction, rather than by a model fitted later to the entire cohort.

Grad-CAM weights convolutional activation maps with gradients of a target score (Selvaraju et al., 2017). We target the originally predicted class and use the last ReLU before the fourth pooling layer. At this point the spatial map is  $16 \times 16$  for a  $128 \times 128$  input. For activation map  $A^k$  and predicted-class logit  $g_c$ , the attribution is

$$L_c = \text{ReLU}\left(\sum_k \alpha_k^c A^k\right), \quad \alpha_k^c = \frac{1}{UV} \sum_{u,v} \frac{\partial g_c}{\partial A_{uv}^k}. \quad (9)$$

For the single output logit  $g$ , the PD target is  $g_c = g$  and the H target is  $g_c = -g$ . The positive map is interpolated to input size and divided by its maximum when nonzero. Heatmap colours express attribution magnitude, independently of the input’s semantic channels.

We calculate Grad-CAM for all 2,985 held-out predictions. A separate occlusion analysis replaces

each non-overlapping  $16 \times 16$  input patch with white and measures the decrease in confidence for the original predicted class. Negative changes are clipped to zero for the spatial occlusion map. This analysis covers all 597 predictions in the first repetition.

Figure 11 illustrates correct and incorrect T4 predictions from both diagnosis classes, selected by the rule in its caption before inspecting the explanations. The aggregate analyses cover every task.

## 8.2 Spatial Concentration and Perturbation

We define foreground as pixels with at least one channel below 250, matching the preprocessing. Foreground enrichment is the fraction of attribution inside that mask divided by the fraction of image area occupied by the mask. A uniform nonzero attribution map has enrichment one.

The participant-averaged Grad-CAM enrichment is 1.803, with interval [1.723, 1.884], indicating concentration around the rendered writing. There are 174 all-zero Grad-CAM maps, 5.83% of the predictions. Enrichment is undefined for those maps, which are omitted before aggregation.

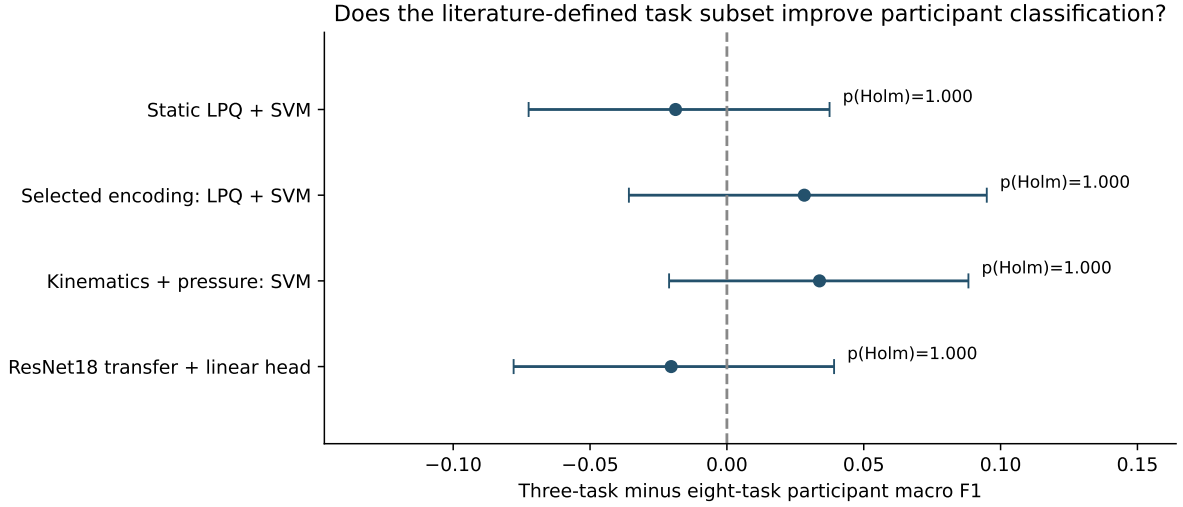


Figure 10: Paired participant-level effects of three-task versus eight-task fusion. The same people contribute to both conditions. Intervals are pointwise 95% participant-bootstrap intervals; Holm correction covers the four methods. For adaptive image methods, changing the task set can also change the training-selected encoding. No task-set contrast survives correction.

To test sensitivity to the highlighted pixels (Figure 12), we whiten the 10% of input pixels with the greatest Grad-CAM values. Confidence in the original class falls by 0.0489 on average, with interval [0.0427, 0.0553]. An equal-area random deletion gives a drop of 0.0033 [0.0016, 0.0050]. The paired advantage of the directed deletion is 0.0456 [0.0395, 0.0518]. Before bootstrapping, repeated predictions are averaged within each participant and task, and tasks are averaged within participant.

Deletion summaries retain all 2,985 predictions, including the 174 all-zero maps. For those maps, the selected pixels follow the ranking routine’s tie handling and carry no attribution-based preference.

The random control matches the number of replaced pixels but not their overlap with foreground or spatial shape. Whitening can also create inputs outside the training distribution. The deletion contrast therefore measures sensitivity under these perturbations and does not fully establish attribution faithfulness.

The two attribution methods show little spatial agreement. Median pixelwise Spearman correlation between Grad-CAM and occlusion is  $-0.0121$  across 540 nonconstant pairs; the remaining 57 correlations are undefined. Concentration on handwriting therefore does not imply agreement about which regions influence the prediction.

We also reinitialize all network weights and compare Grad-CAM for one deterministically chosen held-out image per checkpoint. The 200 checks cover only 11 distinct participants because the selection rule repeatedly chooses the first identifier in each test

partition. The median trained-random correlation is 0.0276 among 160 nonconstant pairs. Low agreement is consistent with dependence on learned weights, but the limited participant coverage and excluded constant maps constrain this check.

### 8.3 Sensitivity to the Encoded Channels

Replacing a whole colour channel with white tests the CNN’s sensitivity to that input component. Across the held-out predictions, the mean confidence drops are 0.0429 for speed, 0.0482 for altitude and 0.0427 for azimuth. Their intervals are [0.0343, 0.0512], [0.0400, 0.0568] and [0.0360, 0.0494], respectively.

These similar global values conceal task variation. Speed has the largest average effect on the spiral, while its effect on T6 is slightly negative. Removing a channel alters colour contrast as well as its coded measurement. The perturbation therefore cannot rank the underlying physiological signals or settle the speed-versus-pressure choice in the earlier SVM experiment.

These checks show sensitivity to writing regions and to all three input channels. They do not identify a disease-specific image feature.

## 9 DISCUSSION

The controlled extensions show how the evidence depends on the encoding, task set and classification endpoint. We examine selection within training, spatially

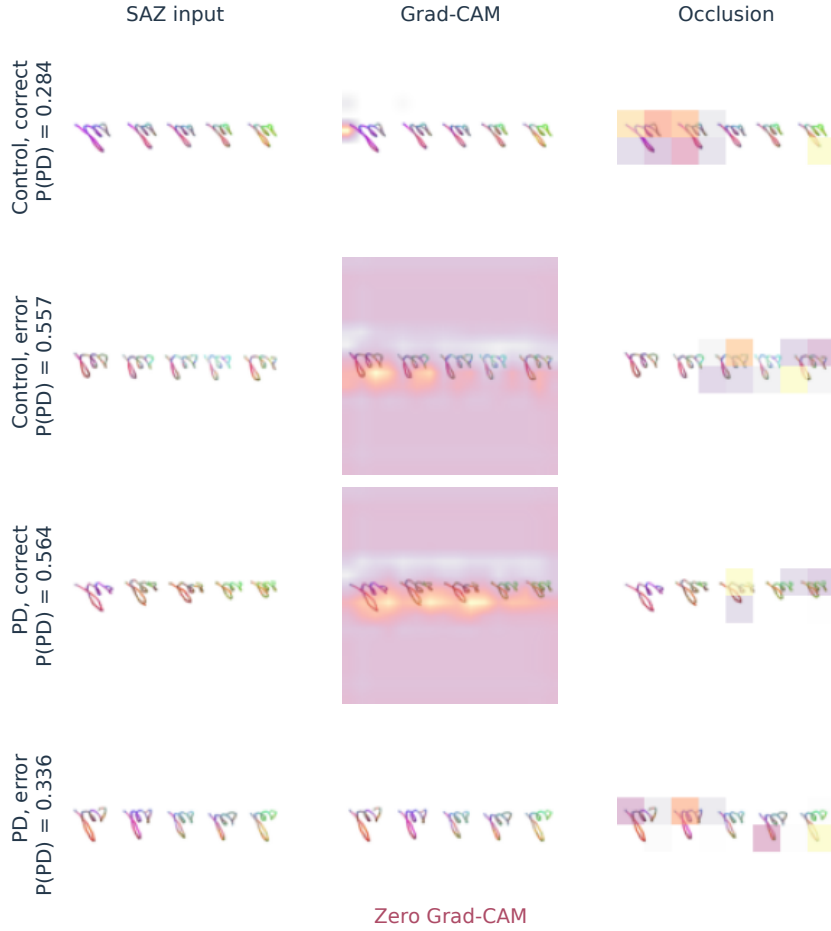


Figure 11: Task-4 explanations for correct and incorrect predictions from both classes. The four cases come from the first repetition, selected at median predicted-class confidence within each diagnosis/outcome stratum. Maps target the original predicted class and use its outer-test checkpoint. Opacity follows per-map attribution. The all-zero Grad-CAM in the PD error is retained. The CNN output is not a calibrated clinical probability.

aligned signals and competitive models before relating the findings to earlier handwriting studies.

## 9.1 Encoding and Spatial Organization

The initial incremental screen found gains on four tasks for *SAZ*, including a 0.0802 increase in spiral macro F1. Its task mean of 0.5752, compared with 0.5643 for static, motivated further evaluation. When encoding selection is confined to training, the nine-encoding procedure achieves 0.5618 and the dynamic-only procedure 0.5643, against 0.5620 for static. None of these paired differences survives Holm correction. These estimates concern adaptive procedures; changes to the inner partitions also distinguish them from the original fixed-*SAZ* screen.

The encoding choices are unstable: *SAZ* is selected in four of the 25 outer training sets in the

dynamic-only search, and no encoding predominates. With fusion rules selected internally for both policies, nine-encoding selection scores 0.5991 against 0.6171 for static. This compares encoding policies under the same fusion-selection procedure; it does not isolate the benefit of selecting a fusion rule. The experiments provide limited support for a generally preferred dynamic encoding on this cohort.

Aligned dynamics score numerically above circular displacement and joint permutation by 0.0082 and 0.0201 in task-mean macro F1. The latter contrast has a positive pointwise interval but an adjusted  $p = 0.1848$ ; neither survives the declared correction. The controls preserve trajectory geometry and joint signal vectors within strokes. Permutation also disrupts local continuity and image texture, so the observed difference cannot identify a physiological mechanism.

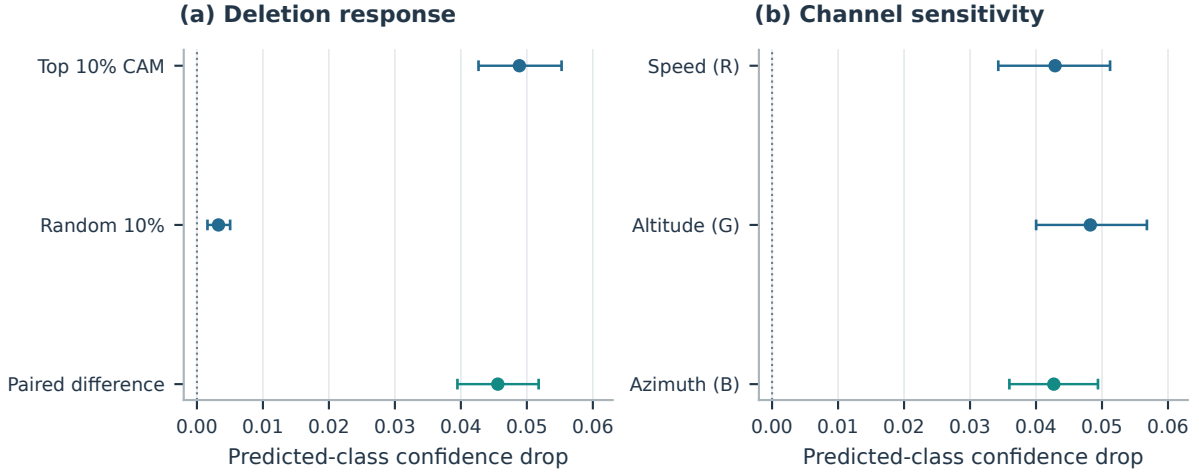


Figure 12: Confidence changes for the originally predicted class. Left: deleting the top 10% Grad-CAM pixels versus equal-area random deletion. Right: whitening each semantic channel. Intervals resample participants after averaging repeats and tasks. These perturbations measure sensitivity of the frozen CNN; the random control does not match foreground overlap, and channel removal also changes colour contrast.

## 9.2 Task Subsets and Competitive Models

With T2/T3/T4 fixed from the literature and majority voting fixed in advance, the selected-encoding LPQ pipeline achieves participant macro F1 of 0.6449, compared with 0.6143 for static. The paired increase is 0.0307, with a 95% interval of  $[-0.0107, 0.0748]$ . This is the largest observed primary-endpoint score among the four pipelines, although none of the six primary contrasts survives Holm correction.

The three-task LPQ fusion exceeds its eight-task counterpart by 0.0283, but the interval includes zero. Reducing the task set also helps the kinematic baseline numerically and lowers the static and transfer scores; none of these changes survives correction. Using three tasks reduces the number of exercises; a reduction in total assessment time or clinical burden was not evaluated. Task means over three and eight exercises describe different evaluation scopes.

Model rankings depend on the endpoint. Kinematics has the highest observed three-task mean, whereas ResNet18 transfer leads the eight-task mean at 0.5813 against 0.5618 for selected LPQ, without a supported difference. The earlier result for a CNN trained from scratch offers no basis for a general claim that CNNs are weaker. The transfer experiment evaluates a frozen ImageNet backbone with a trained linear head; architecture, resolution and search budget differ across pipelines.

## 9.3 Comparison with Handwriting Studies

The reference thesis (Casademunt González, 2023) develops a crop-based CNN on T2 and extends it to T3/T4. Participants are separated into 62 for training and 13 for testing before crops are generated. The reported F1 concerns the PD class. Table 5 therefore compares accuracy with each evaluation scope stated.

Our selected LPQ accuracies on T2/T3/T4 individually are 54.67/56.00/63.20%, below the corresponding thesis values. Its single holdout, crop sampling and crop-size exploration on the designated test group differ from our repeated evaluation. Our three-task fusion is an additional experiment, not a result reported in the thesis.

Drotár et al. combine T2–T8 and report 81.3% accuracy with kinematic and pressure features (Drotár et al., 2016). Their validation repeats ten-fold splitting ten times; the description does not fully establish whether statistical feature screening is repeated within training. Diaz et al. report 86.67% with enhanced images, pretrained features and five selected tasks (Diaz et al., 2019). Their feature selection is explicitly restricted to training, while the same restriction is not established for task selection.

Our accuracies remain below those published combined-task results. The kinematic baseline retains native signal levels, timing and in-air motion, whereas minimaps normalize size and signal levels and retain contact trajectories. LPQ summarizes local phase without retaining its absolute spatial location. Representation, available information, classifier and

Table 5: Contextual PaHaW accuracy comparison. Published values retain their original protocols. Present rows use  $5 \times 5$  participant CV and fixed majority fusion of T2/T3/T4; encoding and model choices use training data. Different inputs, selection procedures and task sets prevent a direct superiority test.

| Study                         | Input and classifier            | Evaluation scope                     | Accuracy (%)          |
|-------------------------------|---------------------------------|--------------------------------------|-----------------------|
| Casademunt<br>González (2023) | CNN, handwriting crops          | T2 / T3 / T4; 62 train, 13 test      | 69.23 / 61.54 / 69.23 |
| Drotár et al. (2016)          | Kinematics + pressure, SVM      | T2–T8; repeated ten-fold CV          | 81.30                 |
| Diaz et al. (2019)            | Enhanced images and ensemble    | Ten-fold CV; T1/T4/T5/T7/T8 selected | 86.67                 |
| Present experiment            | Static LPQ + SVM                | T2/T3/T4 majority fusion             | 61.60                 |
| Present experiment            | Selected encoding + LPQ + SVM   | T2/T3/T4 majority fusion             | 64.80                 |
| Present experiment            | Kinematics + pressure, SVM      | T2/T3/T4 majority fusion             | 62.93                 |
| Present experiment            | ResNet18 transfer + linear head | T2/T3/T4 majority fusion             | 60.27                 |

evaluation procedure vary together; the comparisons do not isolate the source of the performance gap.

## 9.4 Explanation and Generalization

The XAI analysis concerns only the earlier CNN trained from scratch. Its near-zero Grad-CAM–occlusion correlation limits the consistency of the spatial interpretation. Visually plausible maps establish neither disease-specific regions nor an explanation of the transfer model.

All experiments reuse a small cohort that informed development. Participant separation protects each fitted evaluation, while the uncertainty intervals remain conditional on saved predictions and omit the full history of research decisions. Independent-cohort validation is needed to assess transfer across participants, acquisition conditions and demographic distributions.

## 9.5 Reproducibility

The source package includes stage configurations and code for rendering, LPQ and kinematic extraction, frozen CNN transfer, selection, fusion and XAI. Its execution instructions require the original PaHaW recordings, which are not redistributed. Archived searches, partitions, predictions and checkpoints support independent reconstruction of choices and fusion. Checksums connect code and derived artifacts to their inputs.

## 10 CONCLUSION

We introduced a new formulation of pseudodynamic handwriting minimap signatures and evaluated its construction incrementally, followed by training-only

selection, spatial controls and competitive baselines. The observed responses vary across tasks, with substantial uncertainty about predictive gains. On the literature-defined T2/T3/T4 set, selected LPQ fusion reaches macro F1 of 0.6449 against 0.6143 for static; the paired improvement remains uncertain. Neither the controls nor the competitive comparisons establish overall superiority after multiplicity correction. The formulation and reproducible evaluation provide a basis for testing which aspects of pen dynamics generalize beyond this development cohort.

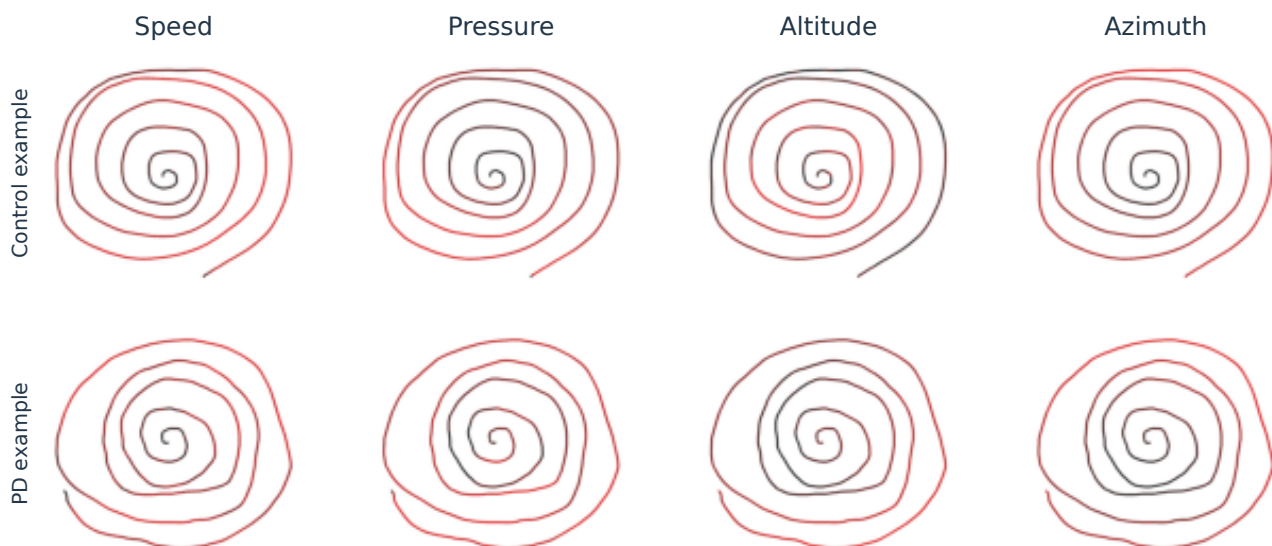
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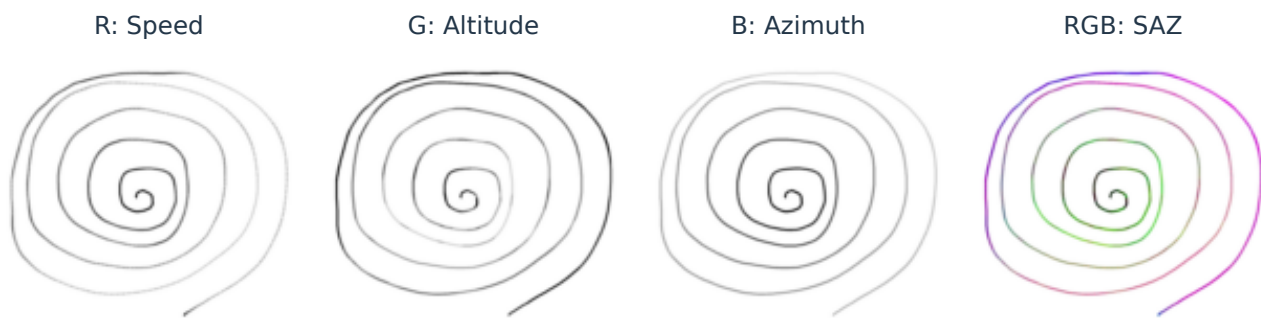
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## A1. Isolated dynamic signals



The same spiral recordings rendered with speed, pressure, altitude or azimuth in the red channel; green and blue remain at 30. The two participants were chosen by a fixed identifier rule, without reference to prediction quality. Scaling uses within-recording fifth and ninety-fifth percentiles of surface-segment values; speed is log-transformed and azimuth is circularly centred.

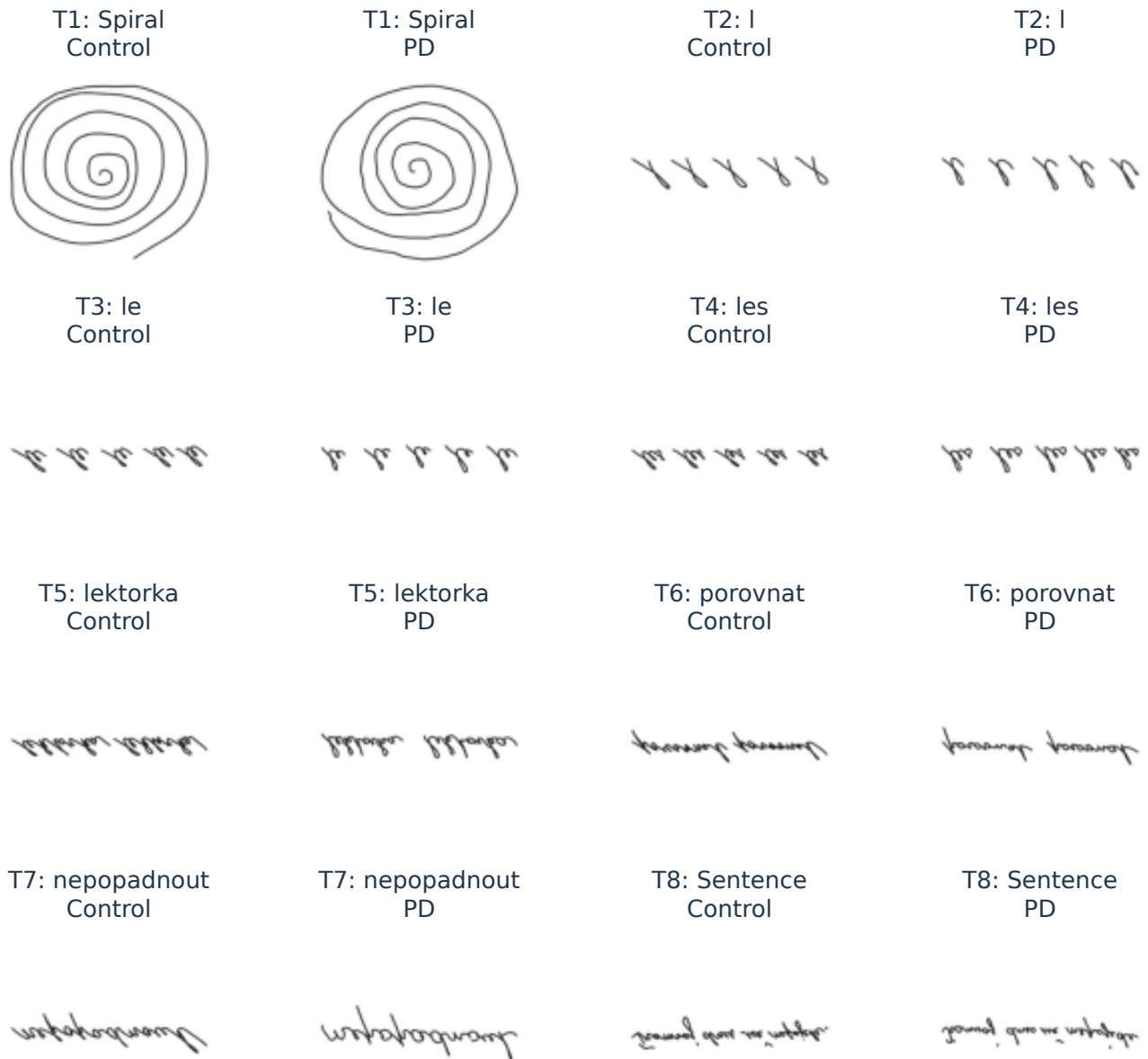
## A2. Reading the RGB representation



Individual channels of a speed–altitude–azimuth (SAZ) spiral, shown in grayscale beside their RGB composite. Red carries normalized log-speed, green altitude, and blue circularly centred azimuth. The grayscale panels display the actual channel intensities, including antialiasing and background. A channel value describes variation within this recording, not an absolute physical value shared across participants.

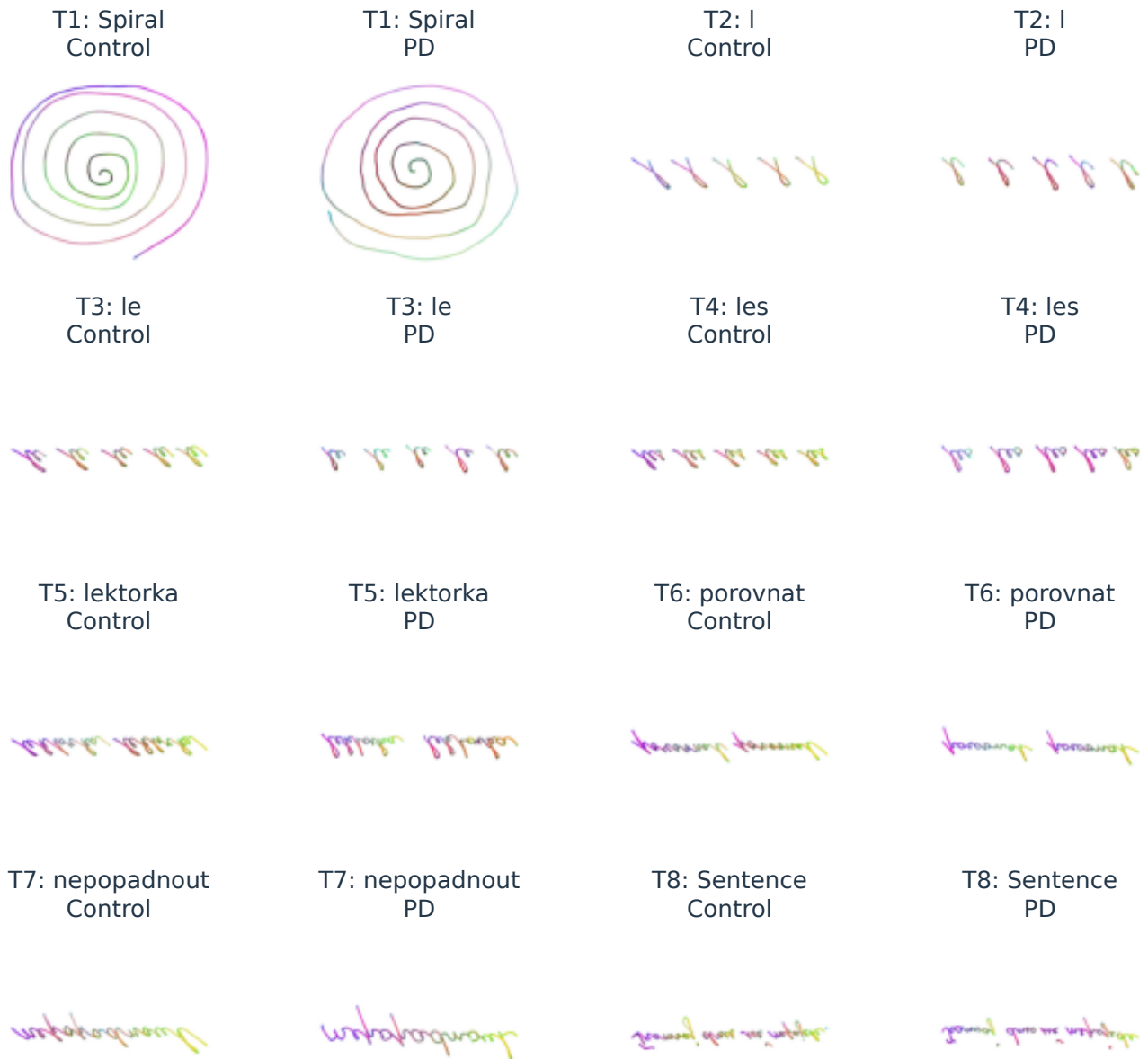
*Editorial option:* Can replace the progressive-encoding figure when explaining channel semantics is the priority.

### A3. The eight tasks as static images



Static reconstructions of all eight PaHaW tasks for the same two illustrative participants, progressing from a spiral through letters and words to a sentence. Each recording is scaled independently while preserving its aspect ratio. The atlas shows the task inventory and within-image geometry; it does not preserve absolute writing size or represent a statistical sample of class differences.

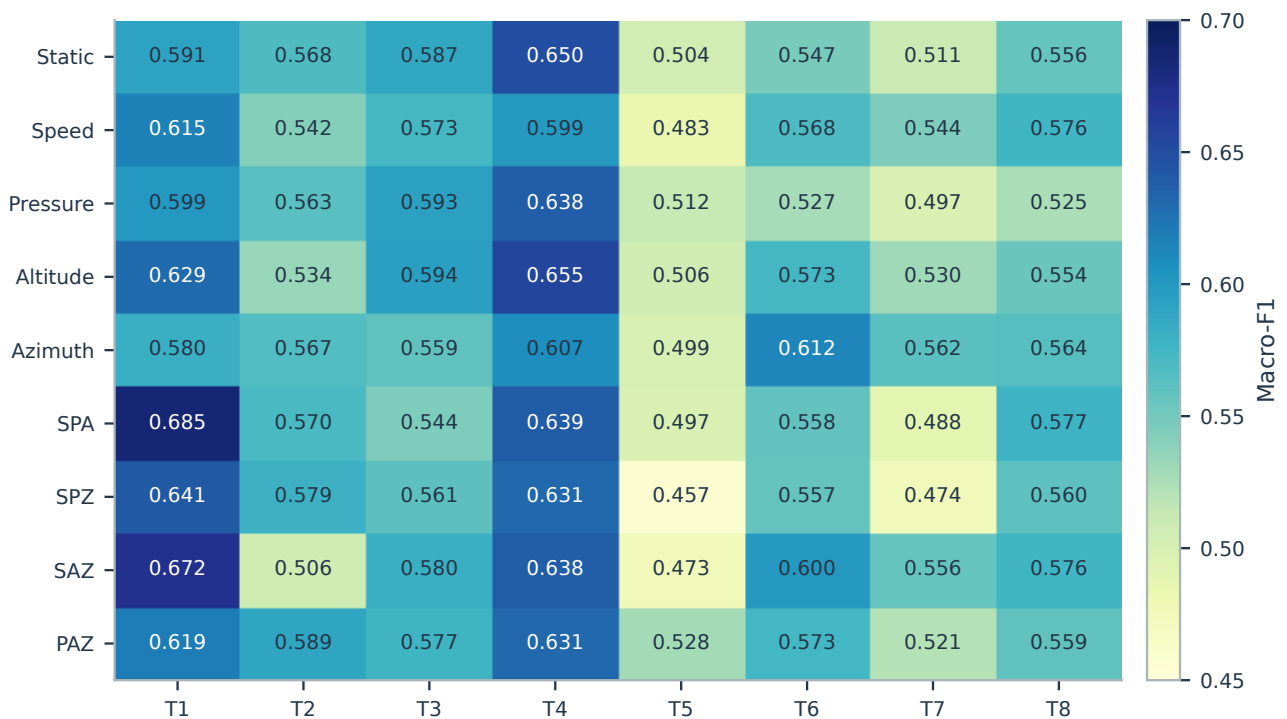
## A4. The eight tasks in pseudodynamic colour



The same task inventory rendered as *SAZ* images for the same two participants as A3. Each task retains its contact-only trajectory and has its own within-recording signal scales. Colours therefore expose local variation without providing an absolute speed, inclination or orientation comparison between recordings. These examples illustrate the encoding rather than evidence of a diagnostic group difference.

*Editorial option:* Can replace the progressive-encoding panel with a broader view of how the representation behaves across tasks.

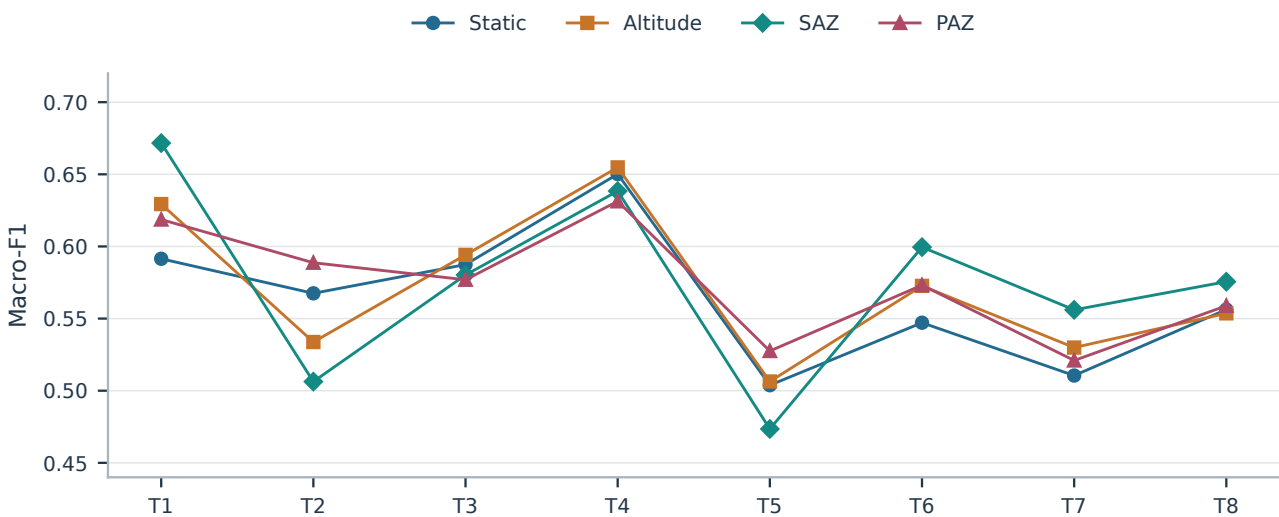
A5. Encoding performance across tasks



Mean task-specific macro F1 for all nine encodings with LPQ-RGB and SVM. Each cell averages the five metrics computed separately from the out-of-fold predictions of each repetition. The common colour scale reveals variation across encodings and tasks. Individual cells are descriptive; their relative shading does not establish a significant pairwise difference.

*Editorial option:* Can replace the encoding-effect figure if task heterogeneity is the focus; retain the paired uncertainty estimates in the text.

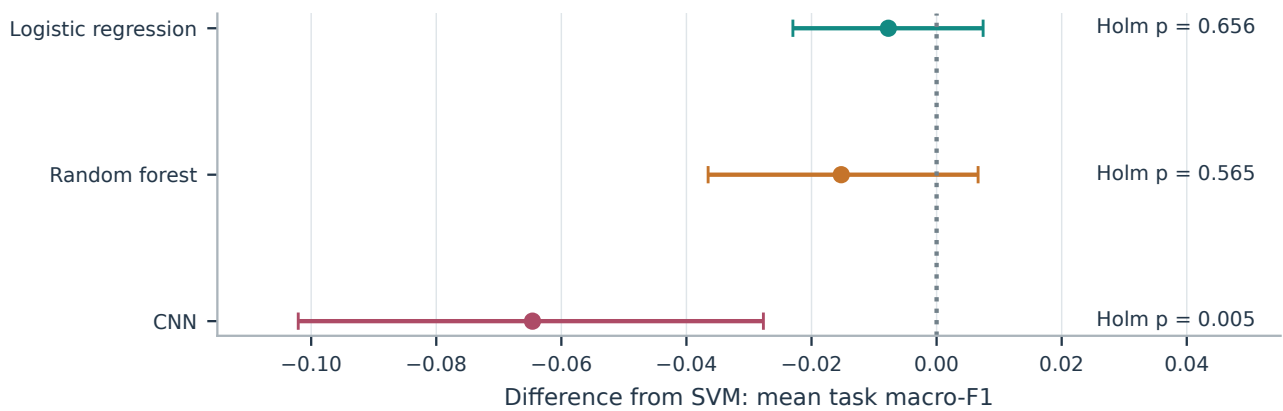
A6. Selected encoding profiles



Task-specific macro F1 for static reconstruction, altitude alone, *SAZ* and pressure–altitude–azimuth (*PAZ*), evaluated under the same SVM protocol. Each value averages five out-of-fold repetition metrics. Lines connect task identifiers to aid comparison; they imply neither a continuous task scale nor a fitted trend. These four profiles are a descriptive subset of the nine evaluated encodings.

*Editorial option:* An alternative to the encoding-effect figure when a small set of task profiles communicates the result more clearly.

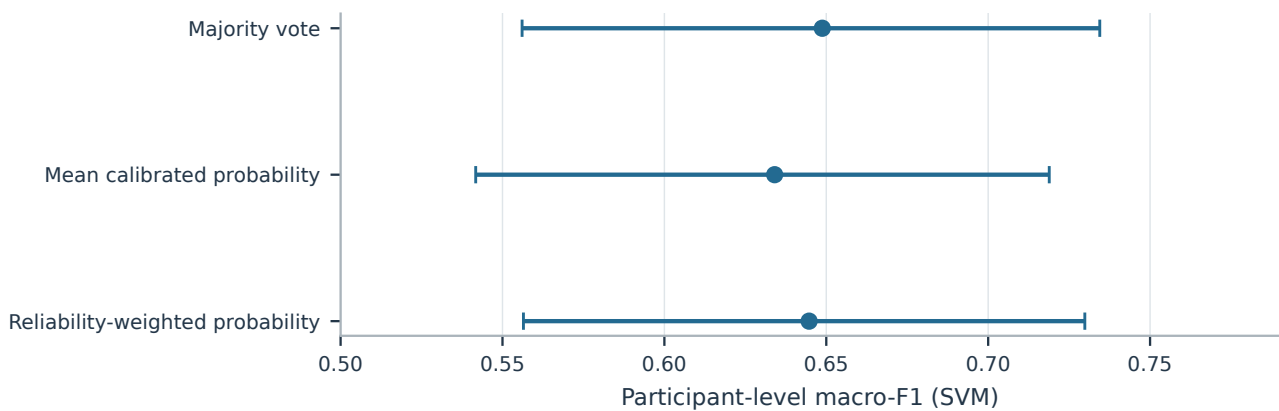
## A7. Classifier differences relative to SVM



Paired differences from SVM in the unweighted mean of task-specific macro F1 across all eight tasks. Intervals are conditional 95% diagnosis-stratified participant-bootstrap intervals, preserving tasks and repetitions together. Labels give Holm-adjusted randomization  $p$ -values for the family of six classifier pairs. Negative differences favour SVM. The comparison is conditional on the selected SAZ representation and saved predictions.

*Editorial option:* Can replace the classifier-profile figure when effect sizes and uncertainty are more important than per-task detail.

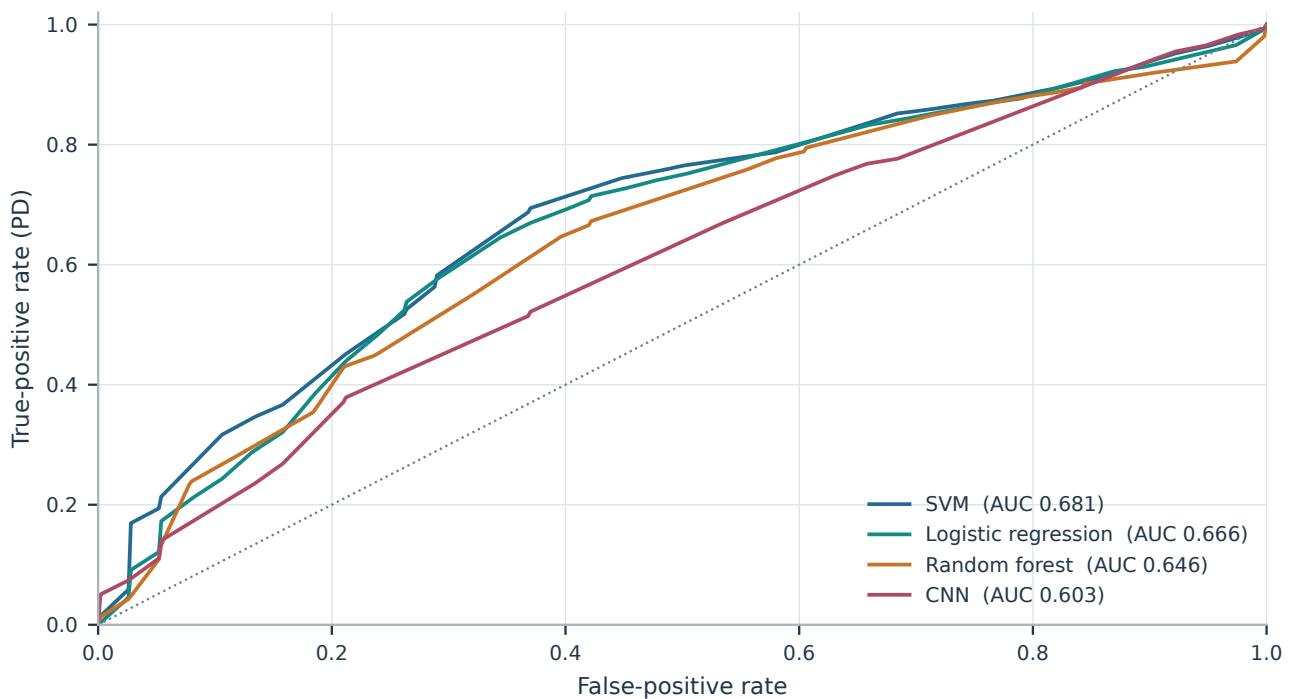
**A8. Three ways to combine SVM task predictions**



Participant-level macro F1 for SVM task-fusion rules, with conditional 95% diagnosis-stratified participant-bootstrap intervals. Calibration and reliability weights use outer-training data only. Mean probability was the primary fusion endpoint; the weighted mean and majority vote were sensitivity analyses. Every rule uses all available tasks, without selecting a subset from outer-test performance.

*Editorial option:* Can replace the participant-fusion comparison when the choice of fusion rule needs its own figure.

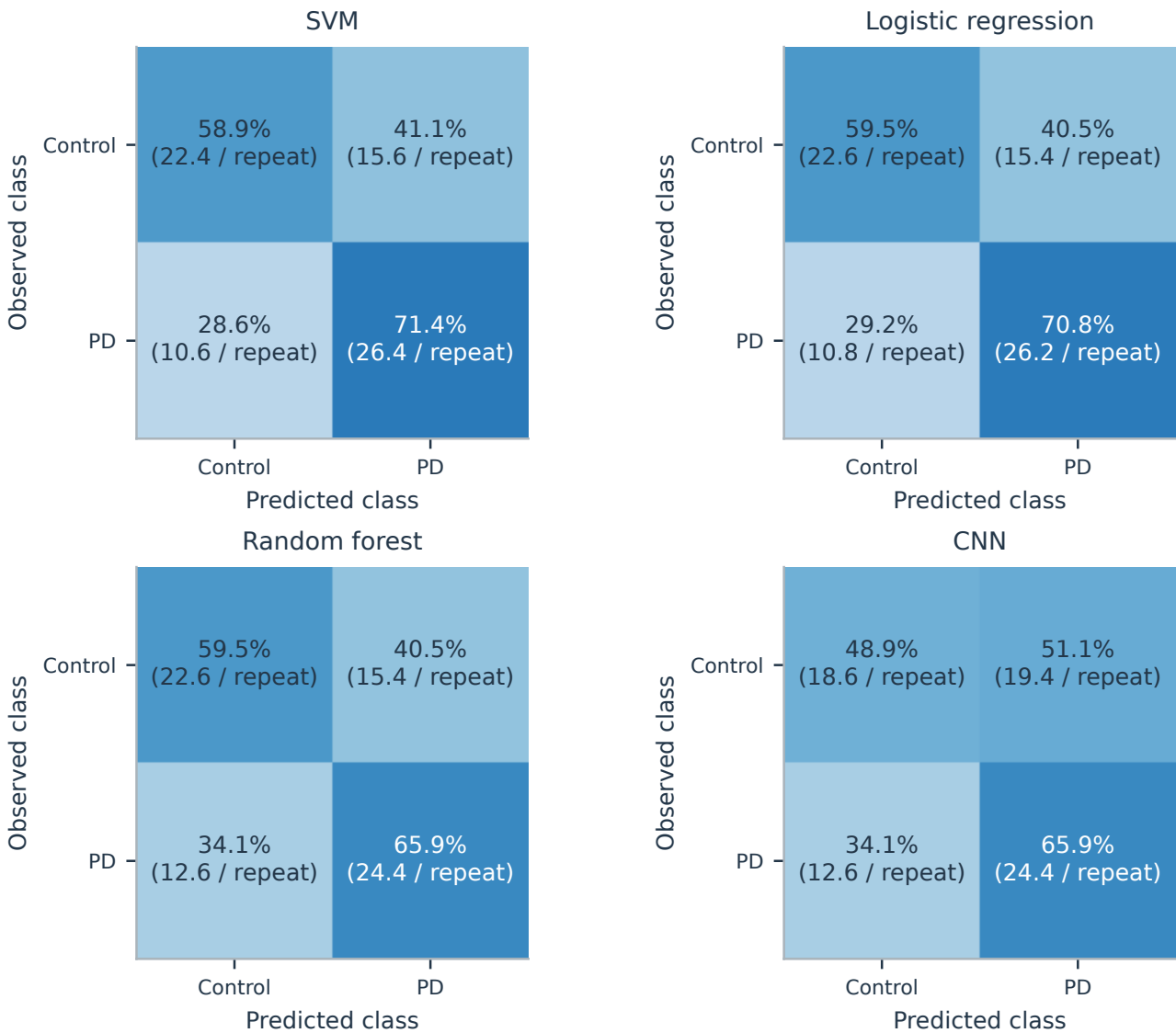
## A9. Discrimination from task-vote fractions



Mean participant-level ROC curves for task-vote fractions. A curve is computed separately for each repetition using 75 distinct participants, interpolated to a common false-positive-rate grid, and then averaged. Legend AUCs average the five original repetition AUCs; they are not computed from pooled repeated predictions. The discrete vote fractions are ranking scores, not calibrated disease probabilities.

*Editorial option:* An alternative to the participant-fusion figure when discrimination over thresholds is the emphasis.

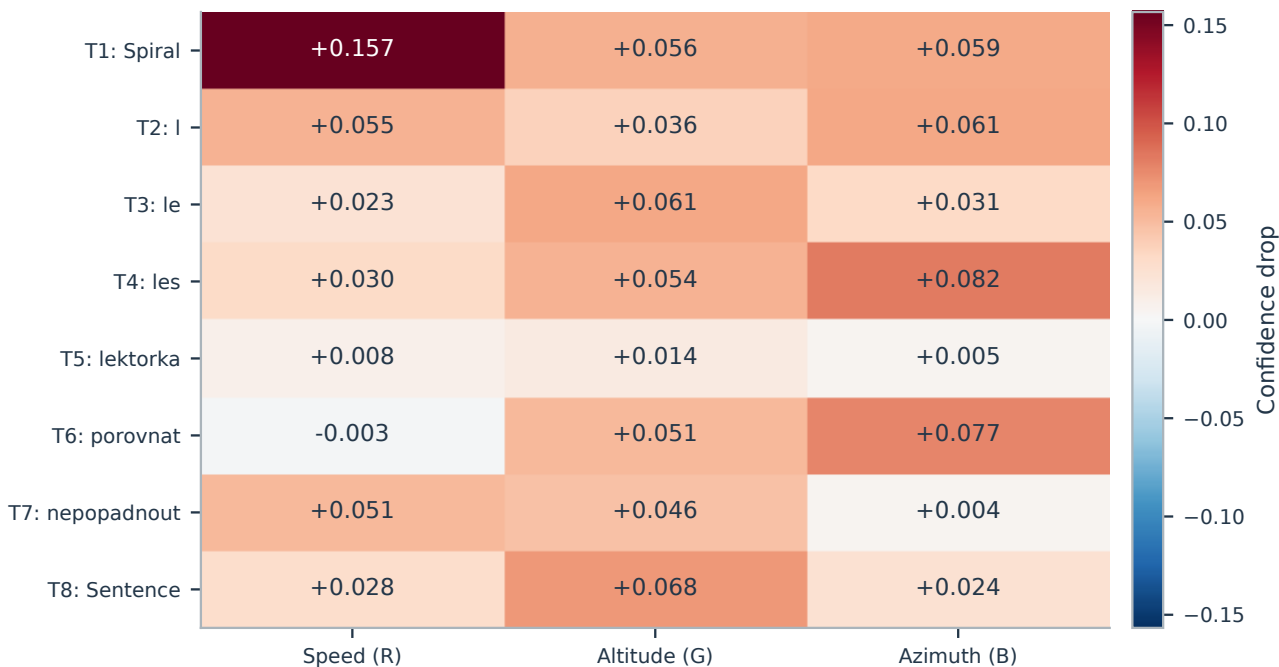
# A10. Errors after task voting



Confusion matrices averaged over five repetitions of participant-level task voting. Each repetition contains 75 distinct participants. Percentages are normalized by the observed class, and parenthetical counts are mean participants per repetition, so fractional counts are expected. Tied task votes are assigned to PD. The repeated predictions are not treated as independent observations.

*Editorial option:* Can replace the participant-fusion figure when the balance of false-positive and false-negative decisions is the main point.

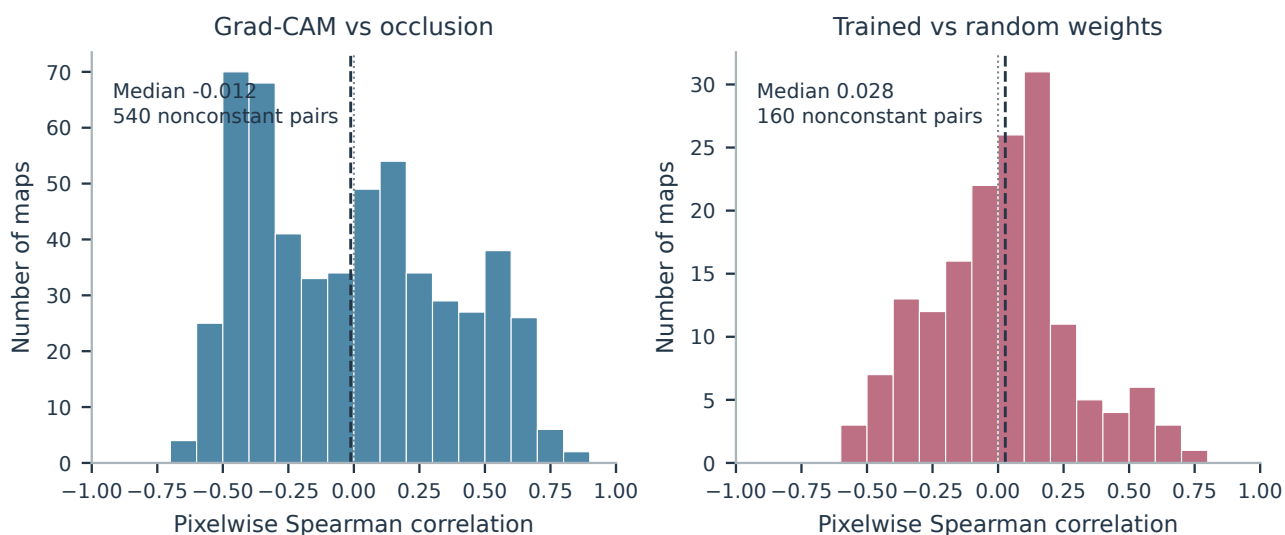
# A11. Channel-removal sensitivity by task



Mean changes in the CNN’s original predicted-class confidence when each complete SAZ channel is replaced with white. Repetitions are averaged within participant and task before calculating task means. Positive values indicate a confidence decrease after removal; negative values indicate an increase. The symmetric colour scale is centred at zero. Channel replacement also changes image contrast, so these effects do not isolate physiological mechanisms.

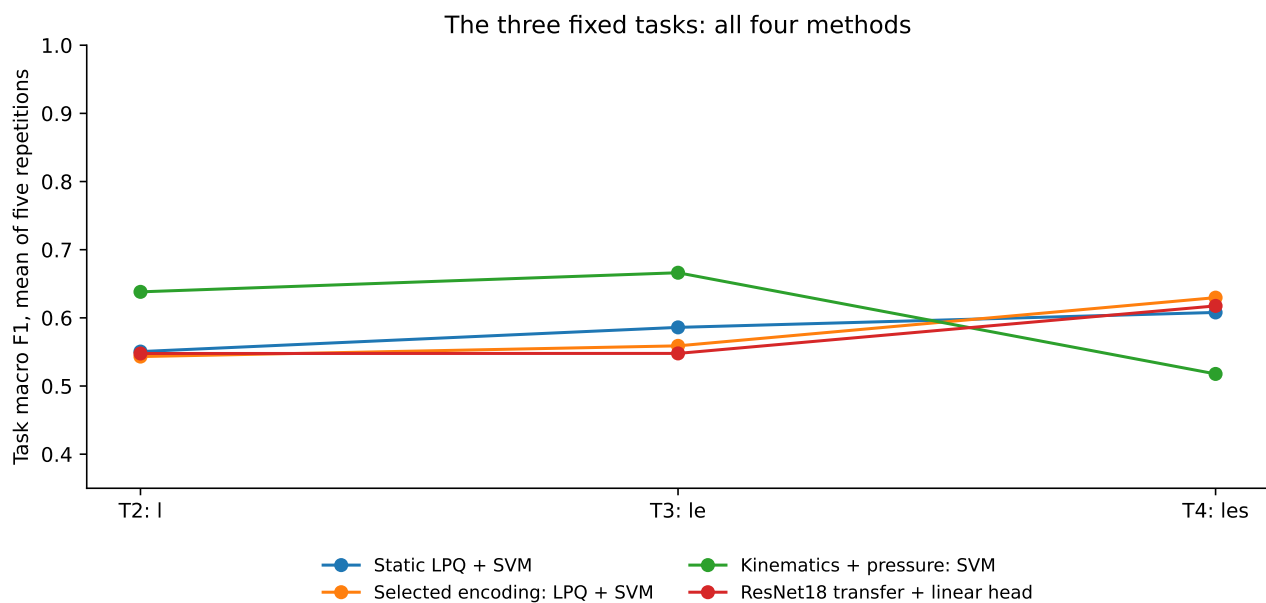
*Editorial option:* Can replace the aggregate channel-removal panel when variation between tasks deserves emphasis.

## A12. Agreement between spatial explanations



Descriptive distributions of pixelwise Spearman correlation. Left: Grad-CAM versus positive occlusion sensitivity for the first repetition, with 540 valid nonconstant pairs among 597 task records. Right: trained versus fully randomized networks, with 160 valid pairs among 200 checkpoint comparisons. Constant maps are omitted. Participants recur across tasks and training sets overlap; these distributions are not independent-sample significance tests.

### A13. Competitive methods on the three fixed tasks



Task-specific macro F1 on T2/T3/T4, the set fixed from the Casademunt thesis before fitting the competitive baselines. Values average five repetitions after pooling participant-separated outer folds. Encodings and classifier parameters are selected within training. The lines aid reading across three categorical tasks; this descriptive panel does not establish task-specific significance.

*Editorial option:* An alternative to the aggregate competitive-method figure when the profiles of the three repeated-pattern tasks are of interest.